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Cambodia University of Technology and Science
Faculty of Sustainable Built Environment
Department of Architecture and Interior Design

Department of Architecture and Interior Design | Graduation Project 2025

GRADUATION PROJECT 2025



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A Water Sensitive Urban Design-Integrated Architectural Response for the Reduction of Urban Flooding at Borey Santepheap II
Shophouse Refurbishment Using Life Cycle Assessment
The Case Study of Riverside Land Integration Public Park Proposal at Chroy Changvar
Beyond the Stalls: Reimagining the Spatial Use of Doeurm Kor Market
Rethinking Rural School Design for Self-Sufficiency
Merging Heritage: Rethinking Airport Hotels for Development Economies in Cambodia
Designing Affordable Housing for Veal Sboav Community Based on 5D BIM Cost Estimation

Department of Architecture and Interior Design
Graduation Project 2025
Bachelor of Science in Architecture

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Publisher's Note

The Graduation Project is established as the culmination of undergraduate studies for the fourth-year architecture and interior design students. It is a problem-based research project that allows the student to identify the problems by doing research and observation, inherent or implicit within students' interests, and delve into specific architectural or interior design or urban design challenges. Through these processes, such as research, analysis, and design development, students could refine their design thinking, gain valuable experience translating concepts into built environments, and prepare for a successful transition into the professional realm.

It is our immense pleasure and pride to congratulate our first-ever cohort of undergraduates on reaching this monumental milestone. During this four-year venture, we truly appreciate all the patience, commitment, discipline, and late nights you endured—experiencing both the sweet and the bitter. The built environment awaits your innovative contributions, creativity, resilience, and vision. Go forth and shape spaces that are not just beautiful but also sustainable and meaningful. Best wishes for your brilliant careers!

Ms. Nou Chanrachna

Head of Department of Architecture and Interior Design

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Monh Neardey
Nath Kananda
Nou Chanracha
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A Water Sensitive Urban Design-Integrated Architectural Response for the Reduction of Urban Flooding at Borey Santepheap II

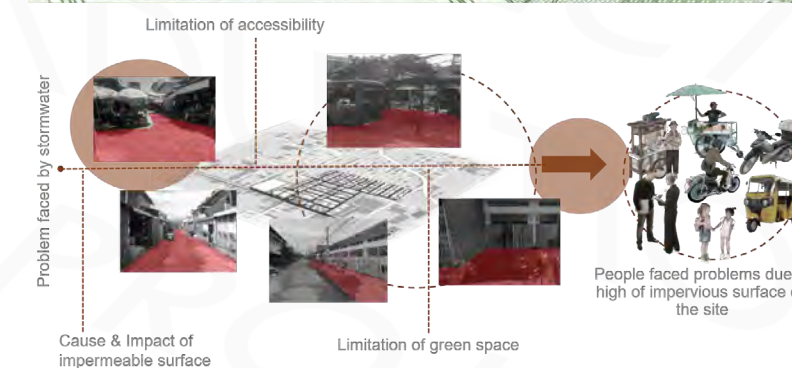
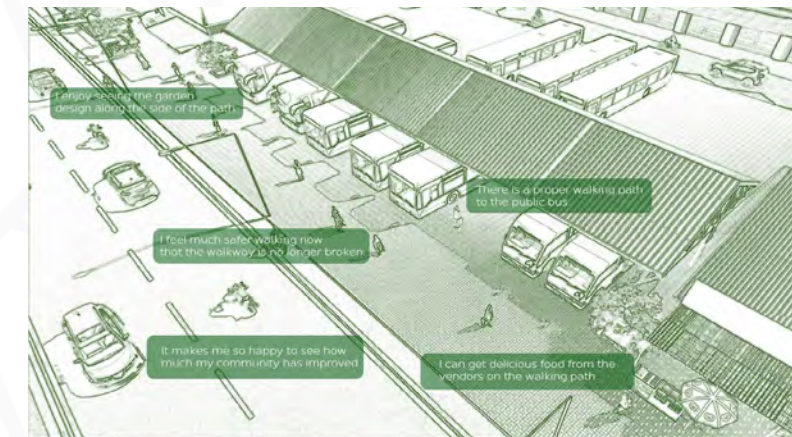
Chhun Vanmarineth

Best Project Poster Award-STRATESA 2025

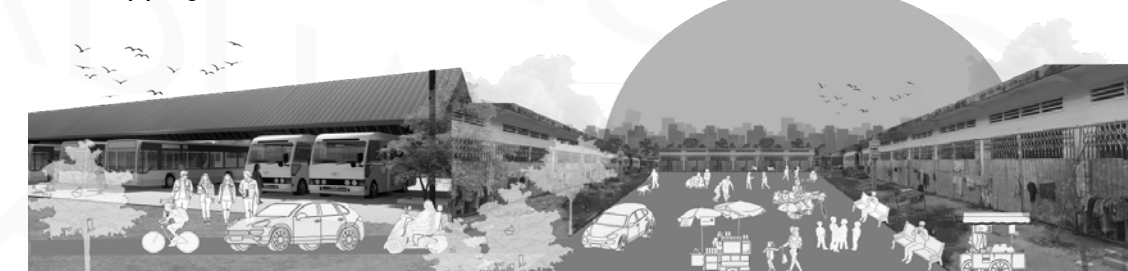
Site Location



Site Problems



Site Mapping

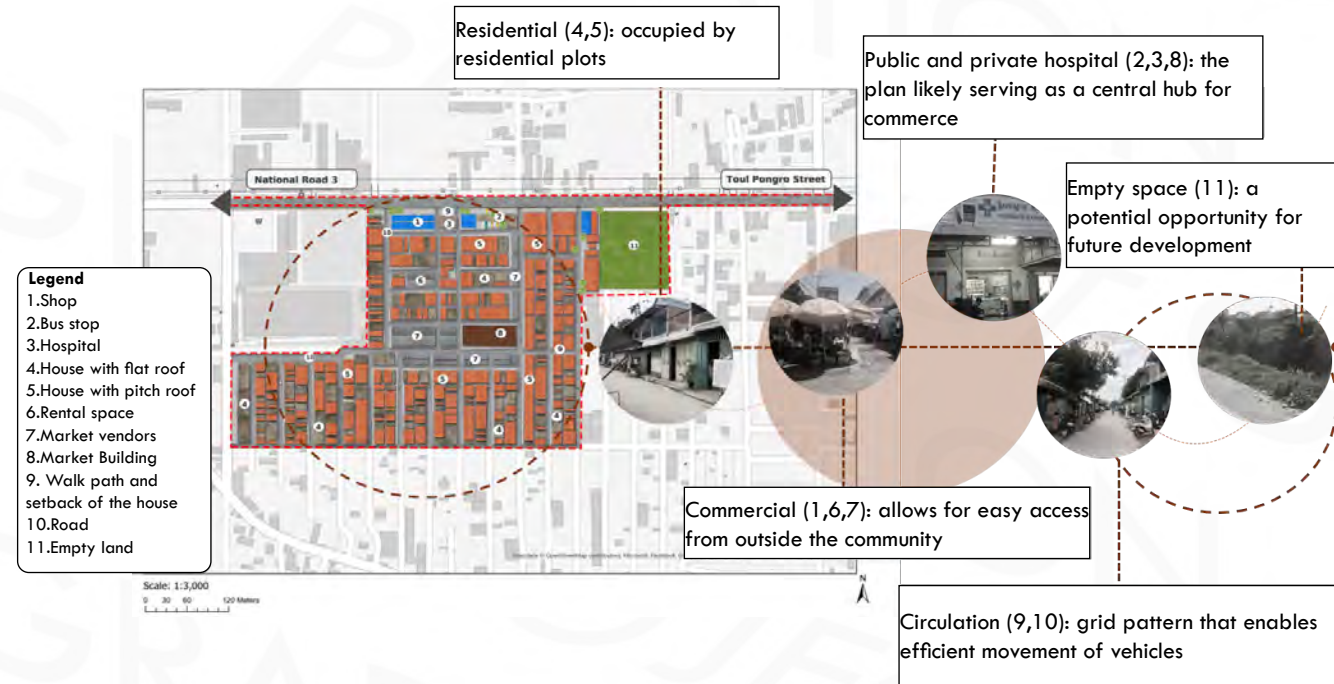


Urban flooding in the residential development of Borey Santepheap II is severe, primarily caused by extensive impervious surfaces that overwhelm the existing drainage system, jeopardizing community safety. This project aims to retrofit the masterplan using Water Sensitive Urban Design (WSUD) for flood mitigation.

The methodology involves land-use analysis followed by applying a responsive architectural framework incorporating four key WSUD elements: vegetated swales, rain garden planter boxes, permeable pavements, and green roofs. The final design presents a comprehensive flood mitigation approach that simultaneously improves community well-being, sustainability, and environmental quality.

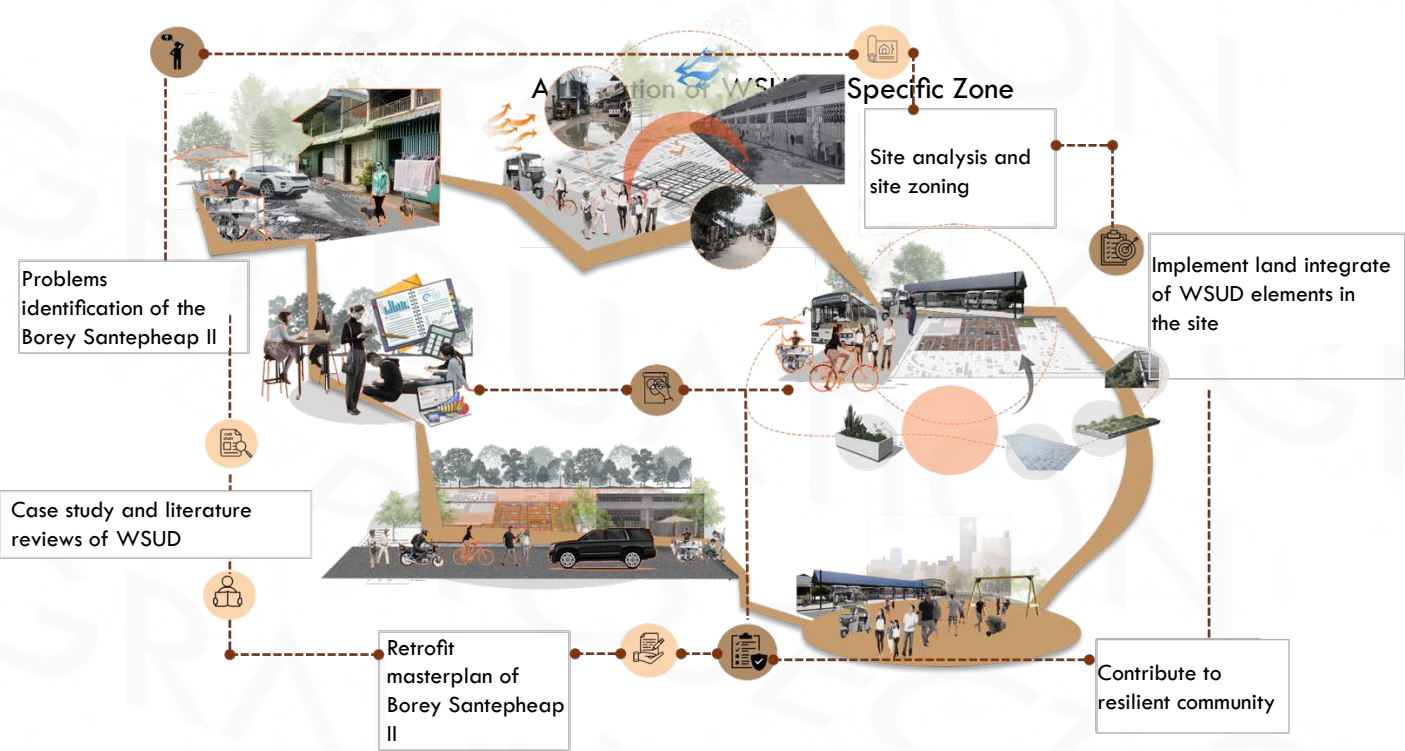
This study demonstrates that an architectural framework integrated with WSUD offers a viable and sustainable alternative to conventional flood management, providing a reproducible model for similar urban expansions.

Existing Master Plan of Borey Santepheap II



Methodology

The methodology begins with a comprehensive site-specific analysis of Borey Santepheap II to identify problem of the site, and the theoretical framework is developed through the water sensitive Urban design (WSUD) literature review, which detail the basic WSUD elements, resulting in retrofit in Borey Santepheap its master plan.



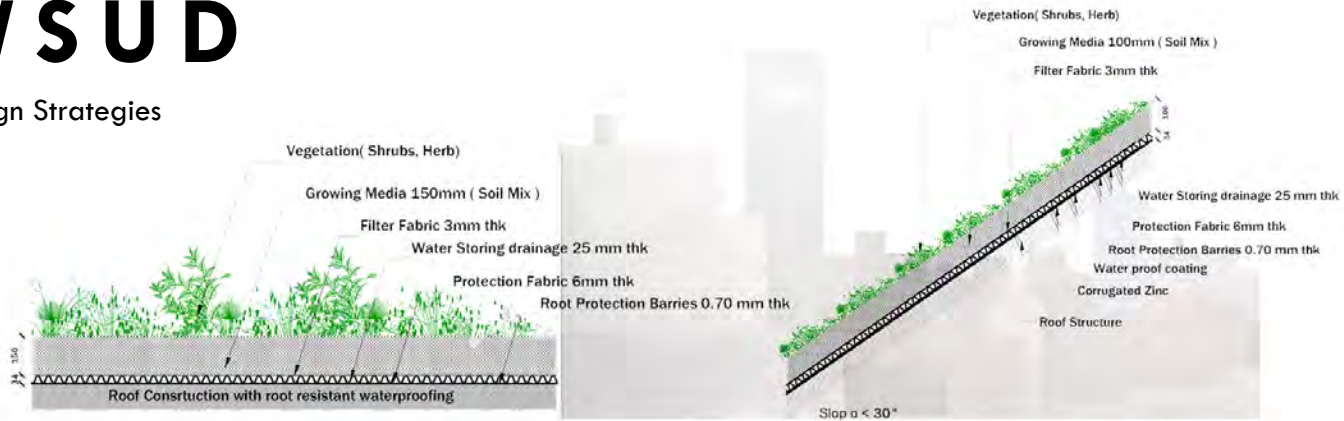
Specifications of the WSUD Elements



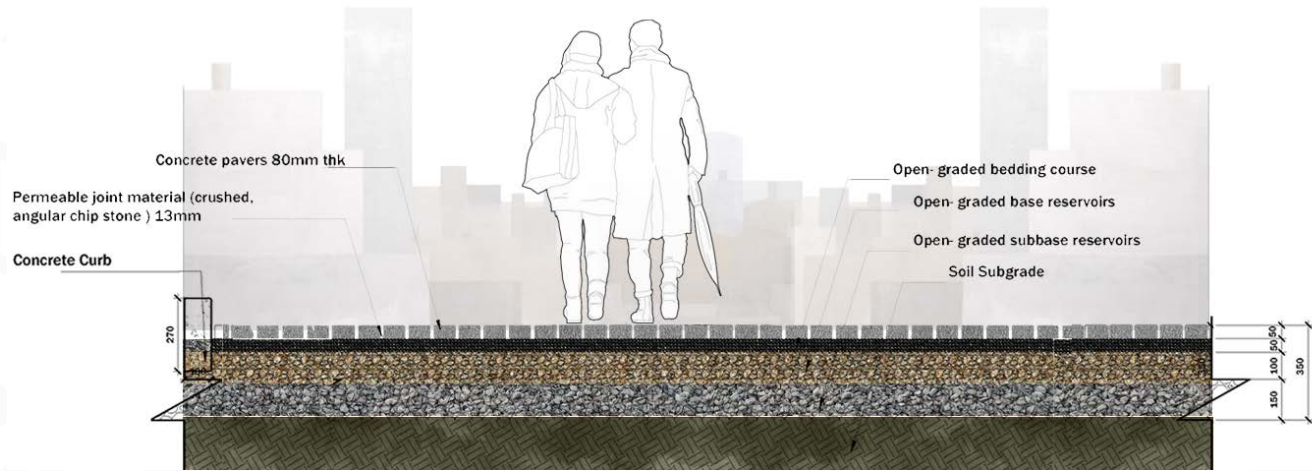
WSUD

Design Strategies

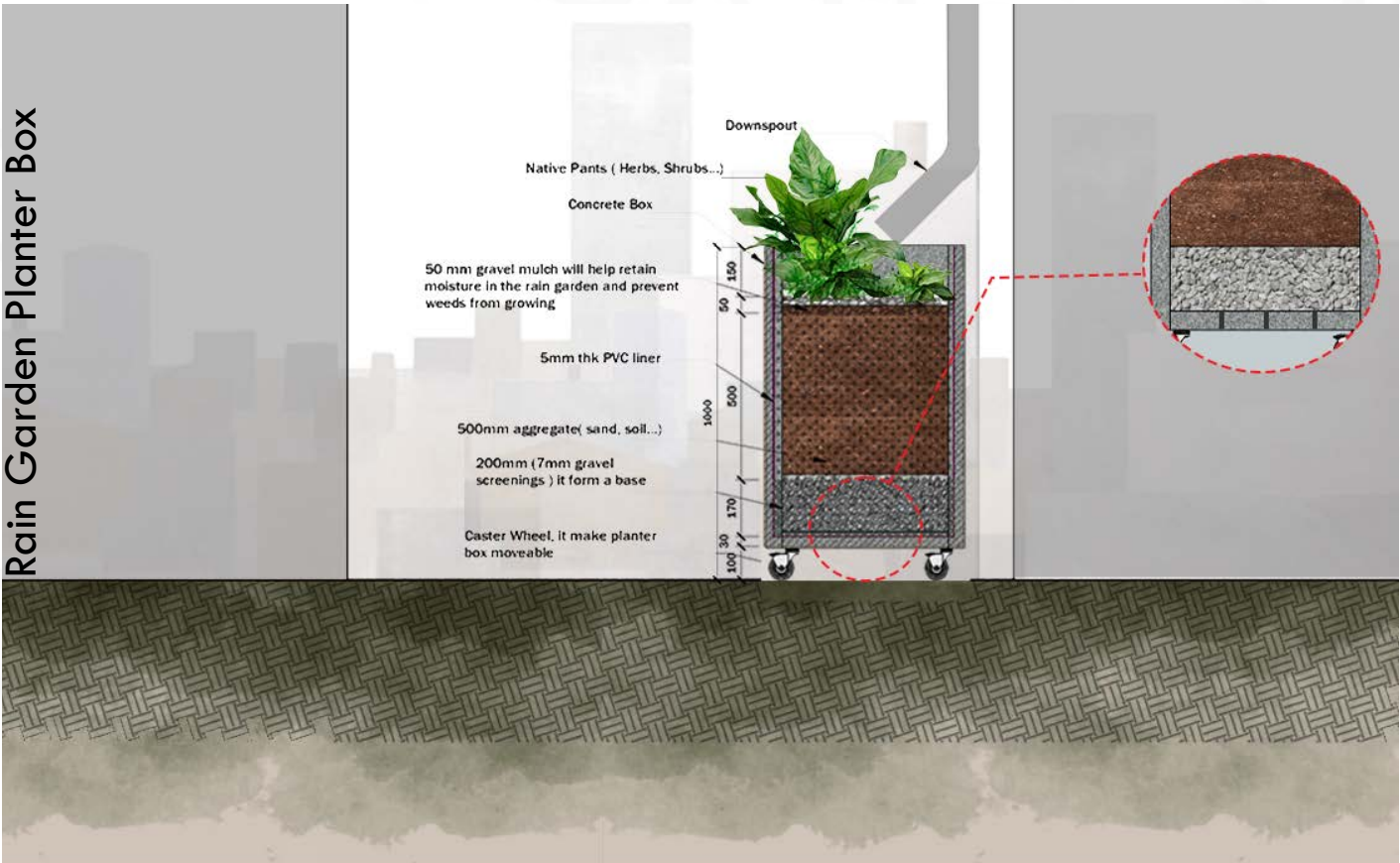
Green Roof



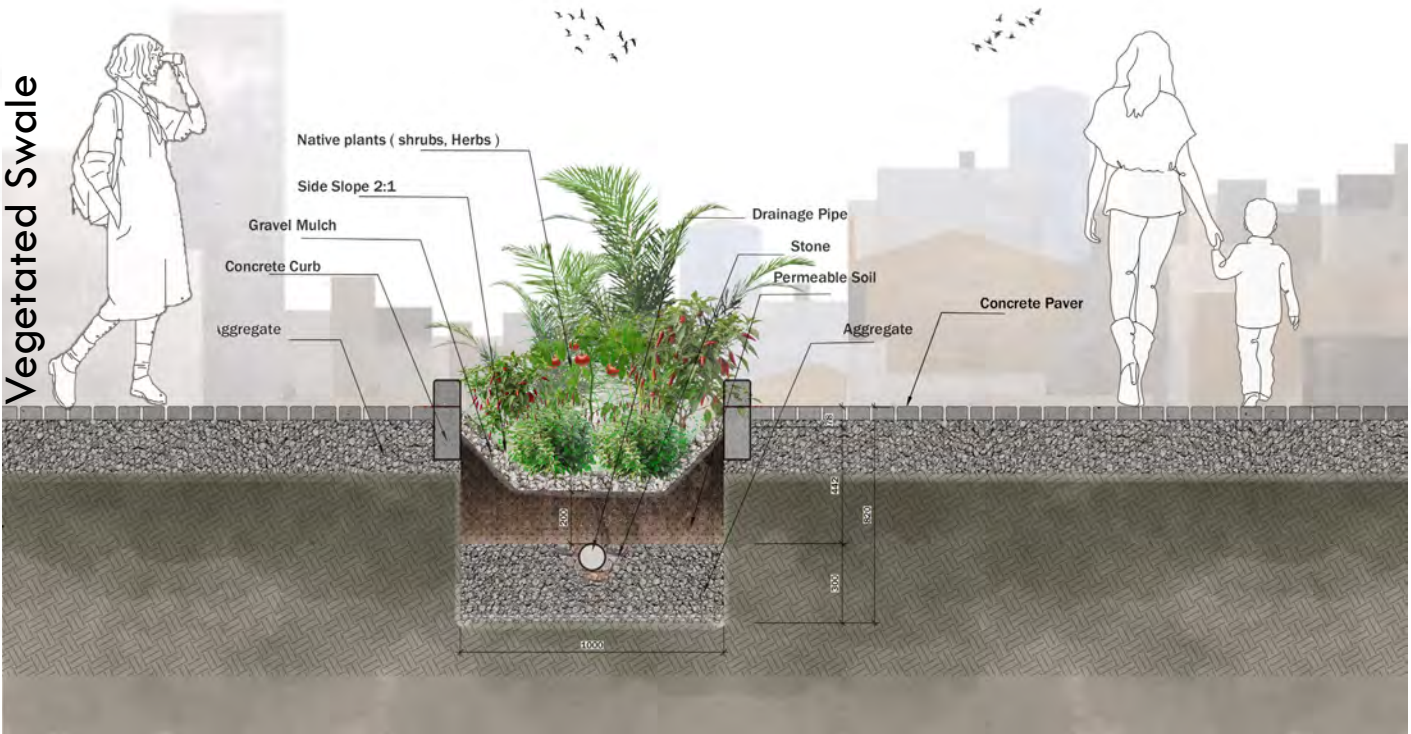
Permeable Pavement



Rain Garden Planter Box

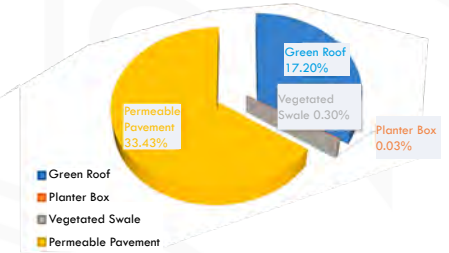


Vegetated Swale



Result of WSUD elements in the specific area

Land Cover	WSUD Element Applied	Percentage of Total Area Transformed
Walkways and roads	Permeable pavements	33.43%
House with flat and pitch roof	Green roof	17.20%
Walkways of bus stop and market	Vegetated swale	0.30%
Setback of the house	Rain garden planter box	0.03%
Total		51 %



Conclusion

This project outlines a retrofit strategy for Borey Santepheap II to mitigate urban flooding in densely populated areas by incorporating Water Sensitive Urban Design (WSUD) elements, including vegetated swales, rain gardens, permeable pavements, and green roofs covering 51% of the site. This design aims to significantly surpass a local benchmark of 48.12% pervious surface, which correlates with a 37.90% reduction in stormwater runoff. Supported by global research, it is concluded that this retrofit can reduce stormwater runoff volume by over 37.90%. This approach not only enhances flood control but also serves as a model for sustainable urban reconstruction and climate adaptation in tropical settings.





Designing Affordable Housing for Veal Sboav Community Based on 5D BIM Cost Estimation

Voun Chhorvan

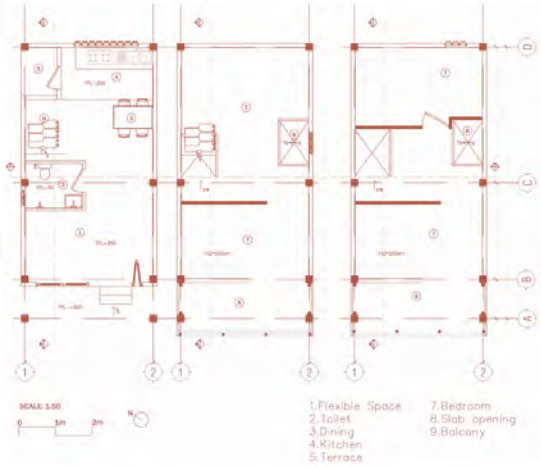


This community, located in Phrek Thum 1 Village, Songkat Kbal Kaoh, Khan Chbar Amboav, Phnom Penh City.



This project explores how five-dimensional Building Information Modeling (5D BIM) can enhance cost estimation and support the design of affordable housing prototypes for the Veal Sbauv community. Two housing models were developed to reflect local living patterns: one for nuclear families and another for extended households. The design process applied Revit for three-dimensional modeling and CYPE software for cost estimation, using real material data such as concrete, timber, roofing sheets, and finishes. Quantities were calculated per prototype to provide accurate cost projections and material planning. The analysis focused on architectural work only, allowing architects to identify the most expensive components and make informed material choices at the design stage. By adjusting these elements with more affordable options, significant savings were achieved.

Floor Plan



The house layout follows the community's needs and respects their living culture. Since most people are used to their current layout, the new design keeps the basic arrangement but makes it more organized and gives more privacy, as they wanted.

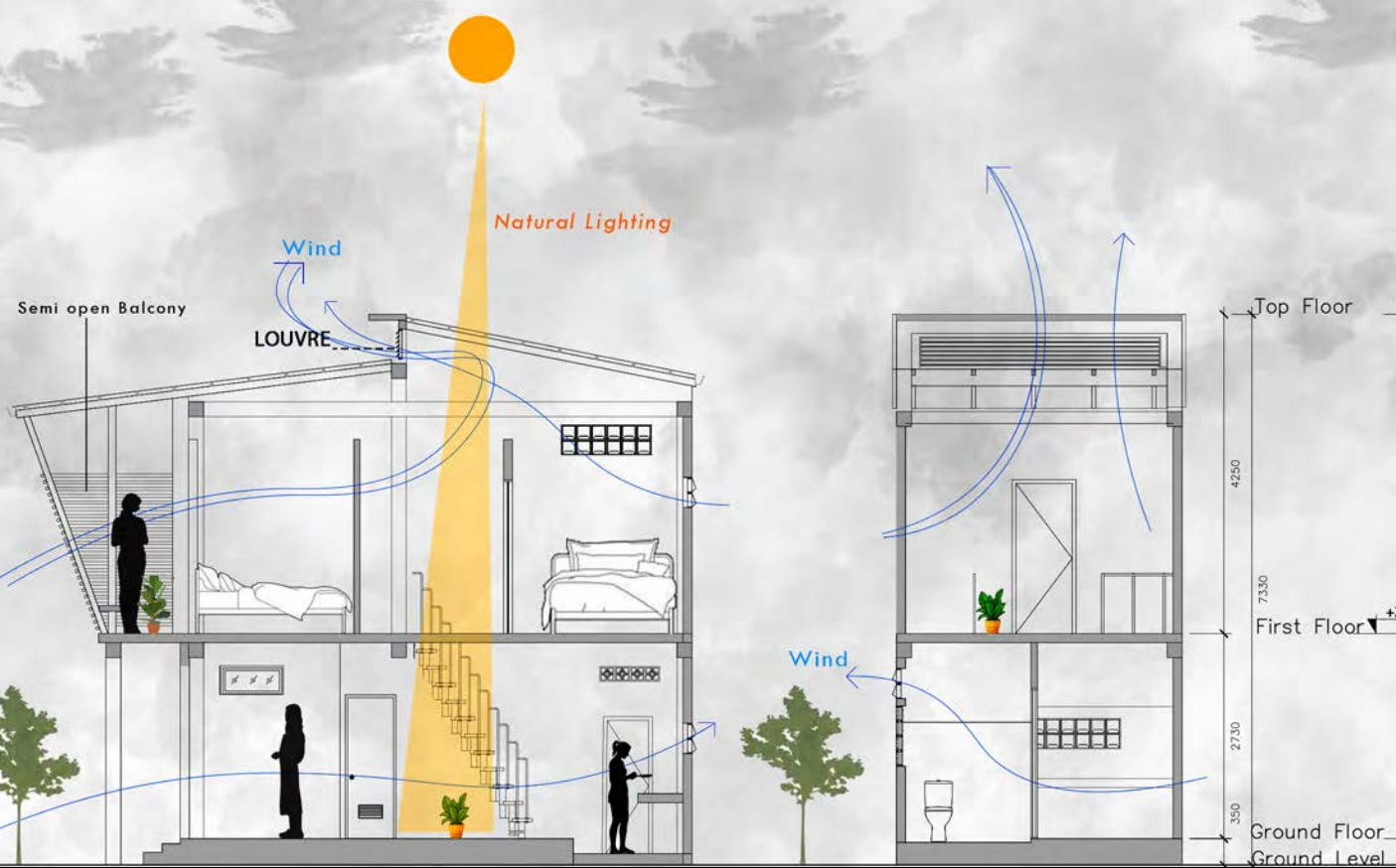


Elevation

This project focuses on three main strategies: space, energy, and materials. For space, the goal is to create layouts that feel comfortable and practical, and respect to the user need.

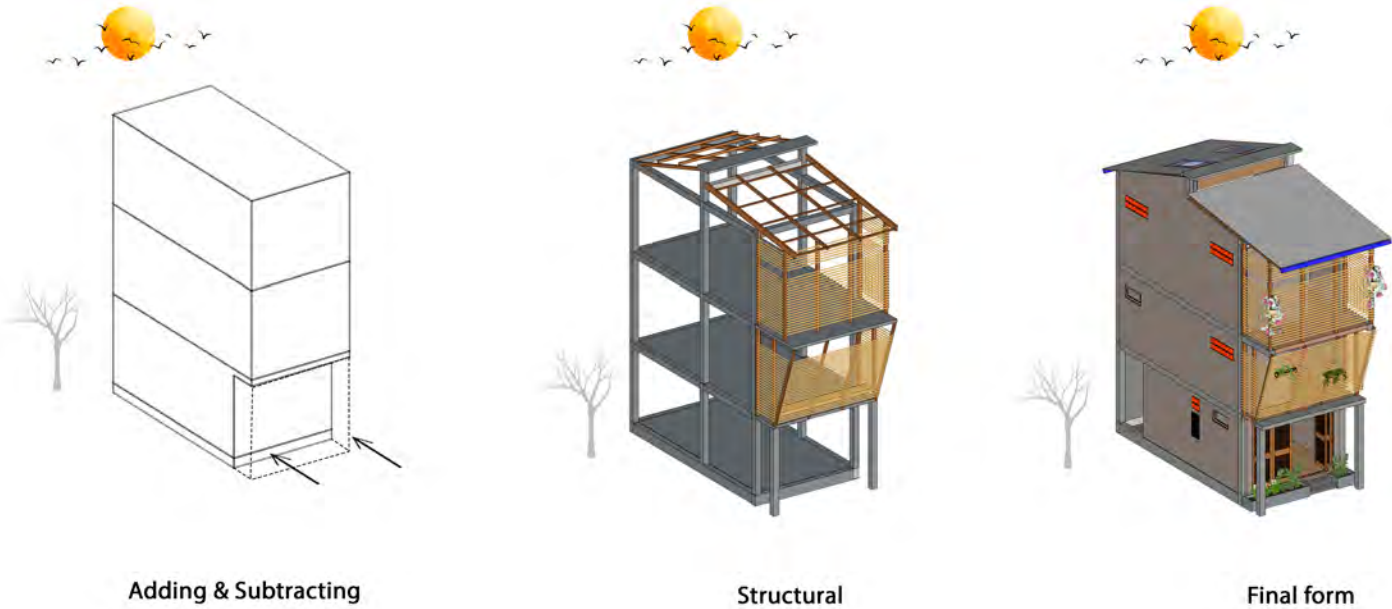


Design Strategies

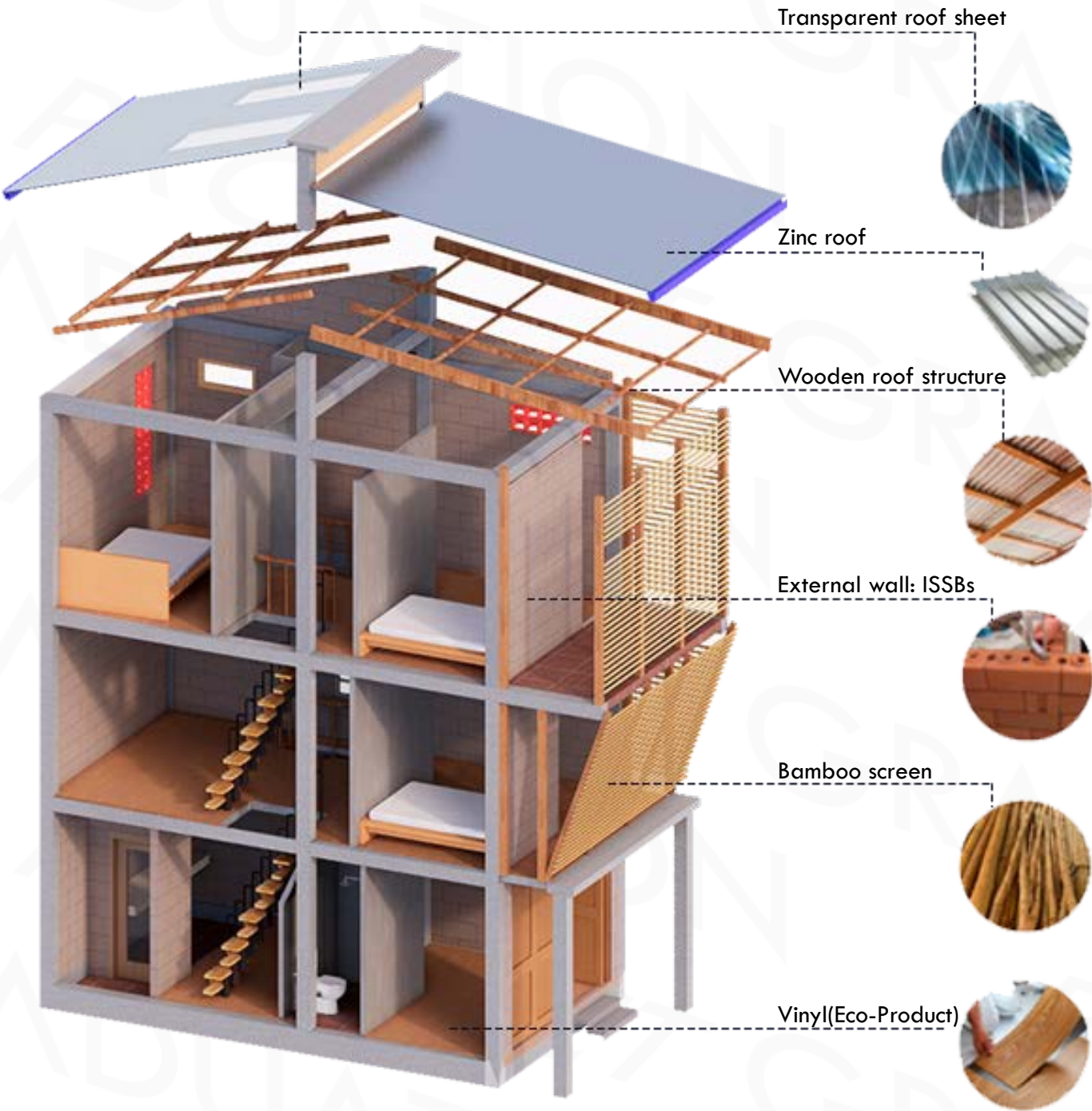


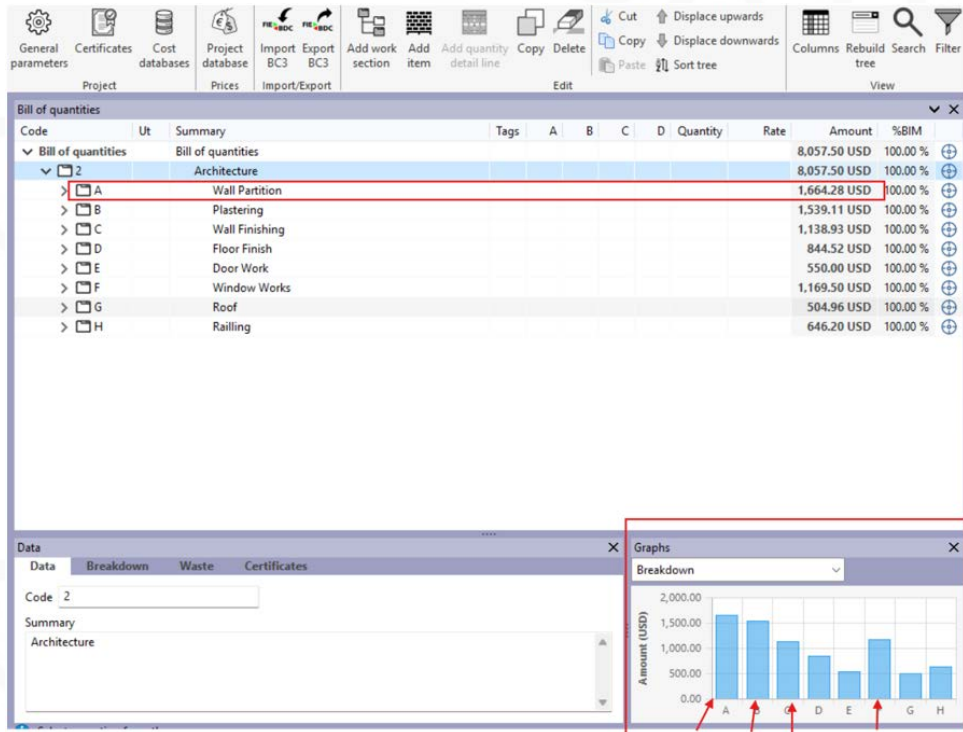
Form Development

The design development process began after conducting both case studies and on-site studies of the community. This included in-depth interviews (Conducted by author, 2025) with community members to understand the challenges they face in daily life, as well as the real needs of the community. The study also documented the conditions of their current housing and gathered other important data to ensure that the design responds directly to people's lived experiences.



Material Selection





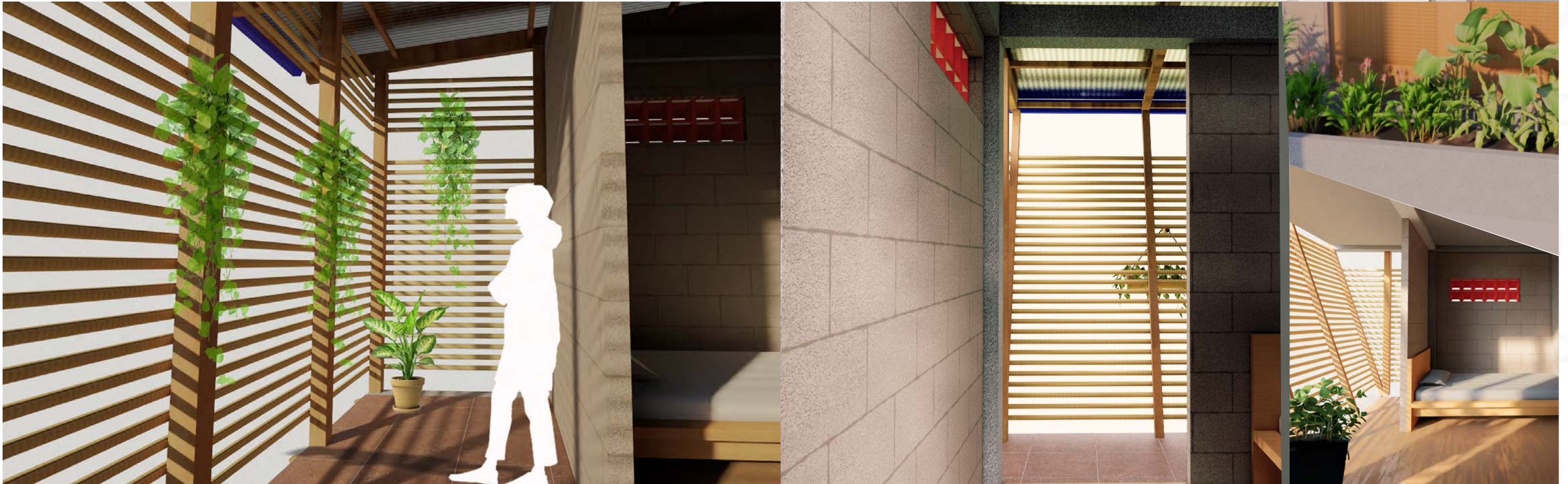
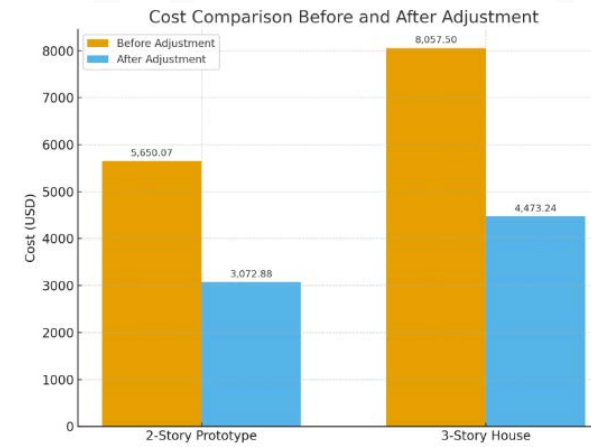
BOQ Visualize in Open BIM

After completing the cost database and applying Open BIM rules of measurement, the cost breakdown shows that wall partitions represent the highest portion of the total cost, followed by roof, plastering, and window works. These elements have the greatest influence on overall project cost. Open BIM enables accurate cost estimation by transferring reliable model data into cost databases, allowing architects to identify expensive components early. This supports cost and material optimization through informed comparison of alternatives and design adjustments. In this project, external walls contribute most to total cost, consistent with studies linking wall materials to socio-economic factors.

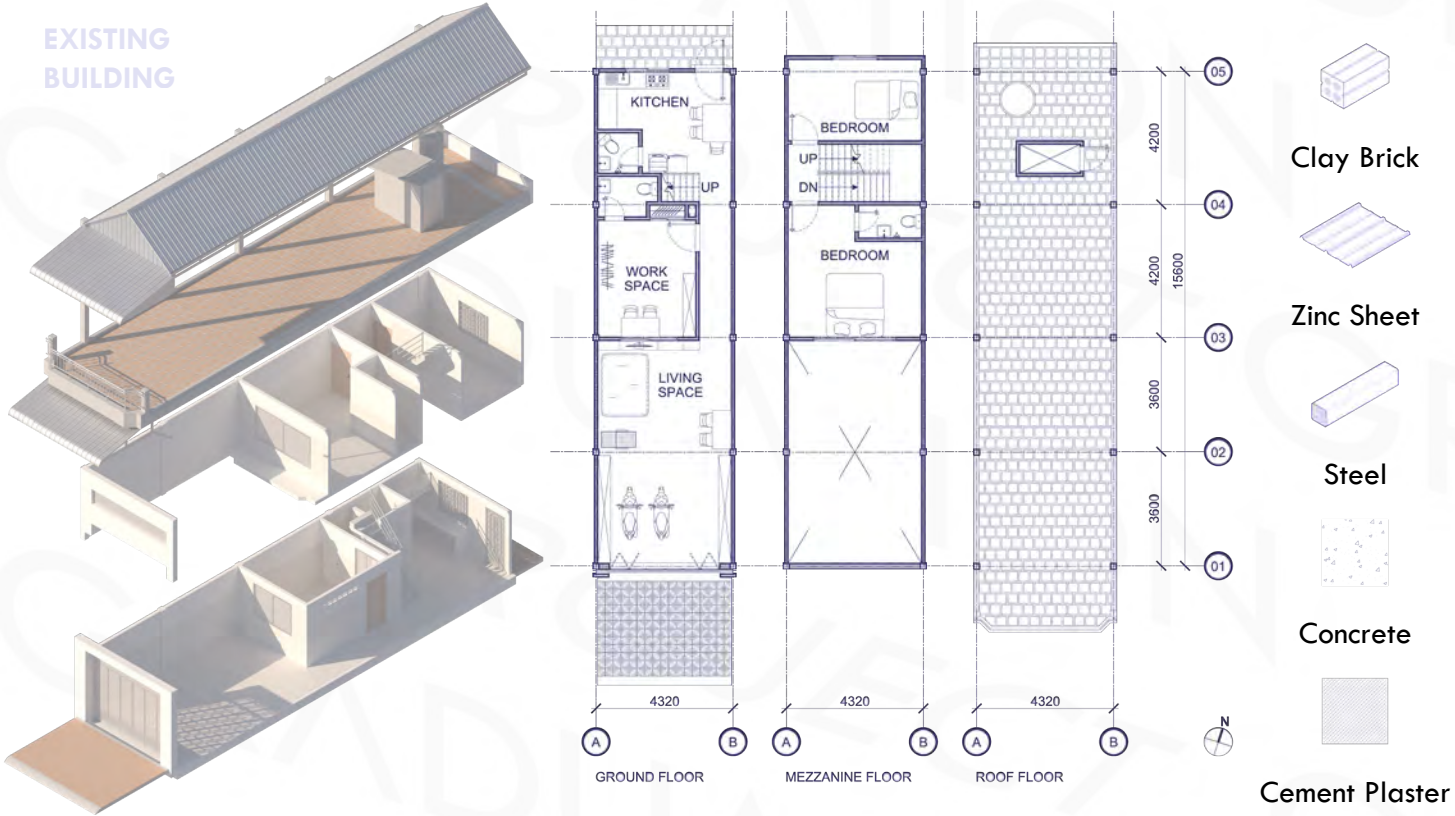


Result & Discussion

By adjusting the highest-cost elements (external walls, roofing, and windows) to more affordable options, significant savings were achieved: the two-story prototype cost decreased from \$5,650.07 to \$3,072.88 (45.61%), while the three-story house dropped from \$8,057.50 to \$4,473.24 (44.48%). These results demonstrate that 5D BIM helps architects make data-driven decisions, optimize material selection, and balance quality with affordability in small-scale affordable housing projects.



EXISTING BUILDING



REFURBISHMENT DESIGN



Shophouse Refurbishment Using Life Cycle Assessment

Hou Khemrakbot

The project is a refurbishment design of a terrace shophouse in Russey Keo, Phnom Penh.

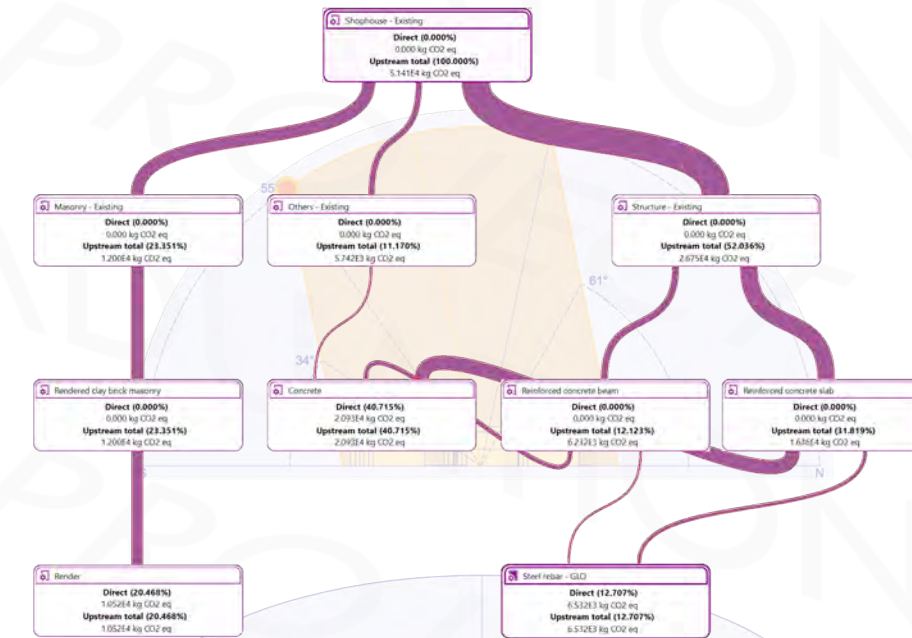
The two main aspects of the project are life cycle assessment (LCA) and sustainable design strategy. The life cycle assessment covers the embodied carbon of construction materials. The design strategies focus on lighting, ventilation, thermal comfort, and greenery.

With the considerations toward indoor environmental quality and environmental impacts, the design aims to convey a pertinent architectural design approach, as well as contribute to the encouragement of sustainable architecture.

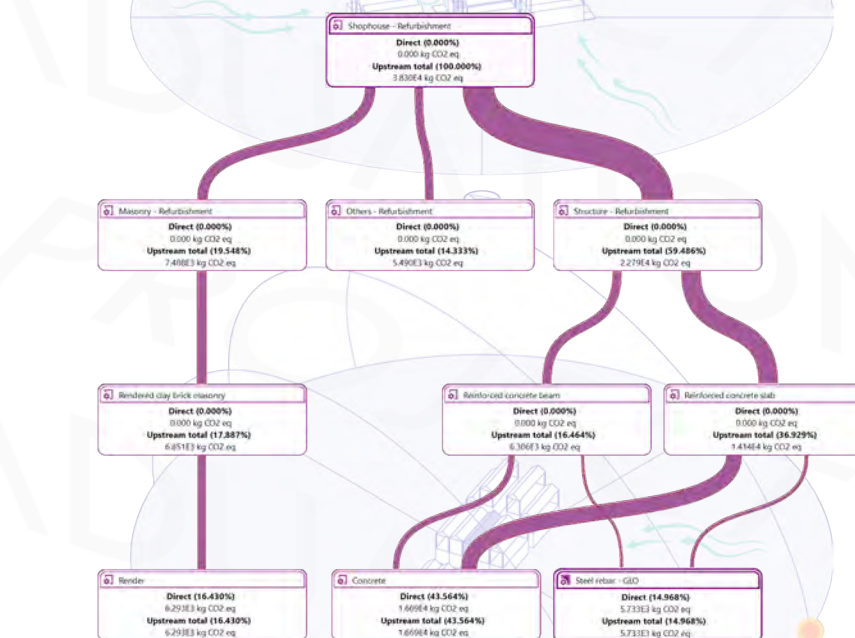
The result of life cycle assessment shows that all impact categories are reduced. The main indicator for result interpretation, global warming, is reduced from 5.16×10^4 kgCO₂e to 3.85×10^4 kgCO₂e, which is roughly a 25% reduction.

GLOBAL WARMING CONTRIBUTION

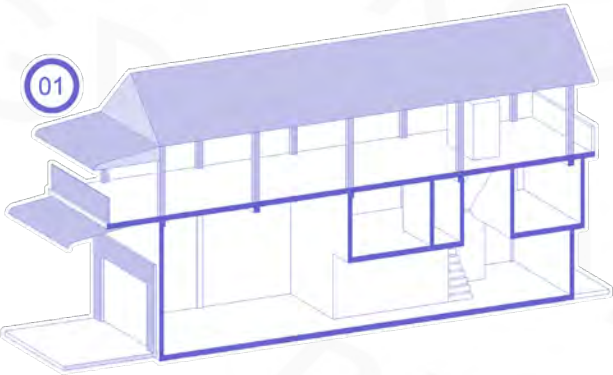
EXISTING BUILDING



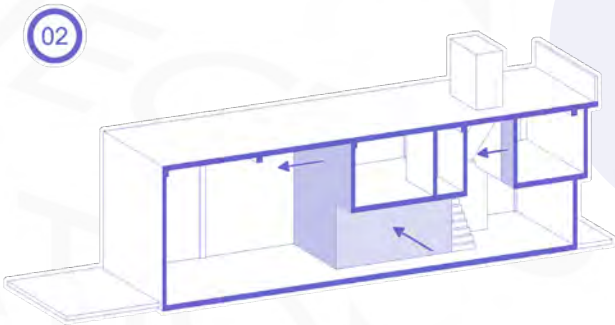
REFURBISHMENT DESIGN



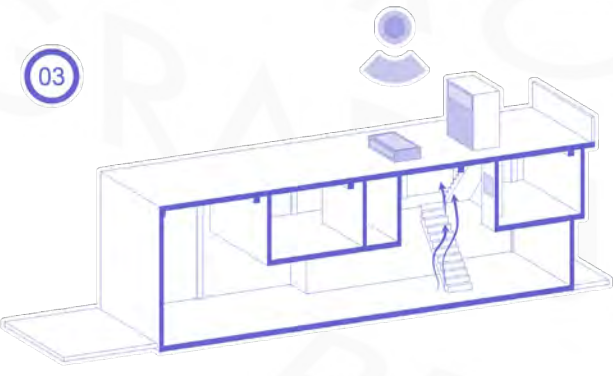
FORM DEVELOPMENT



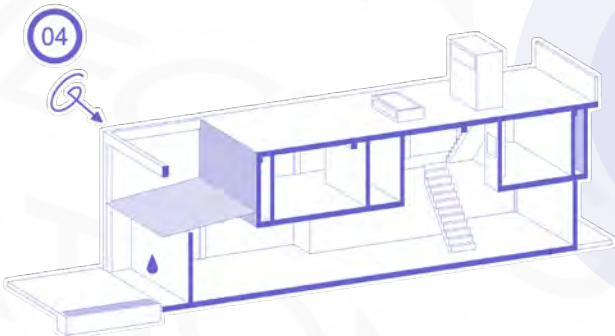
Eliminating Exceeding Elements



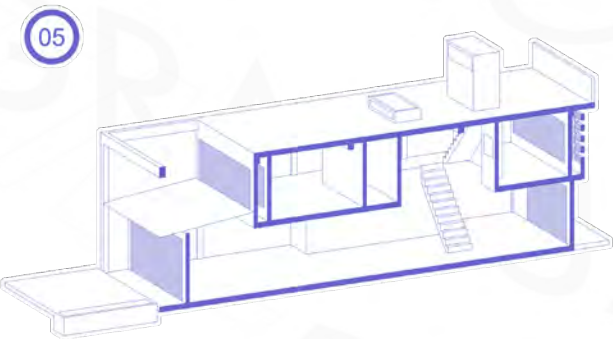
Modifying Interior Volumes



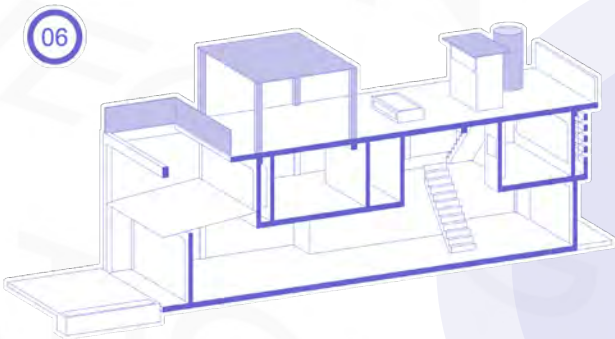
Facilitating Ventilation & Sunlight



Shielding Heat & Collecting Rainwater



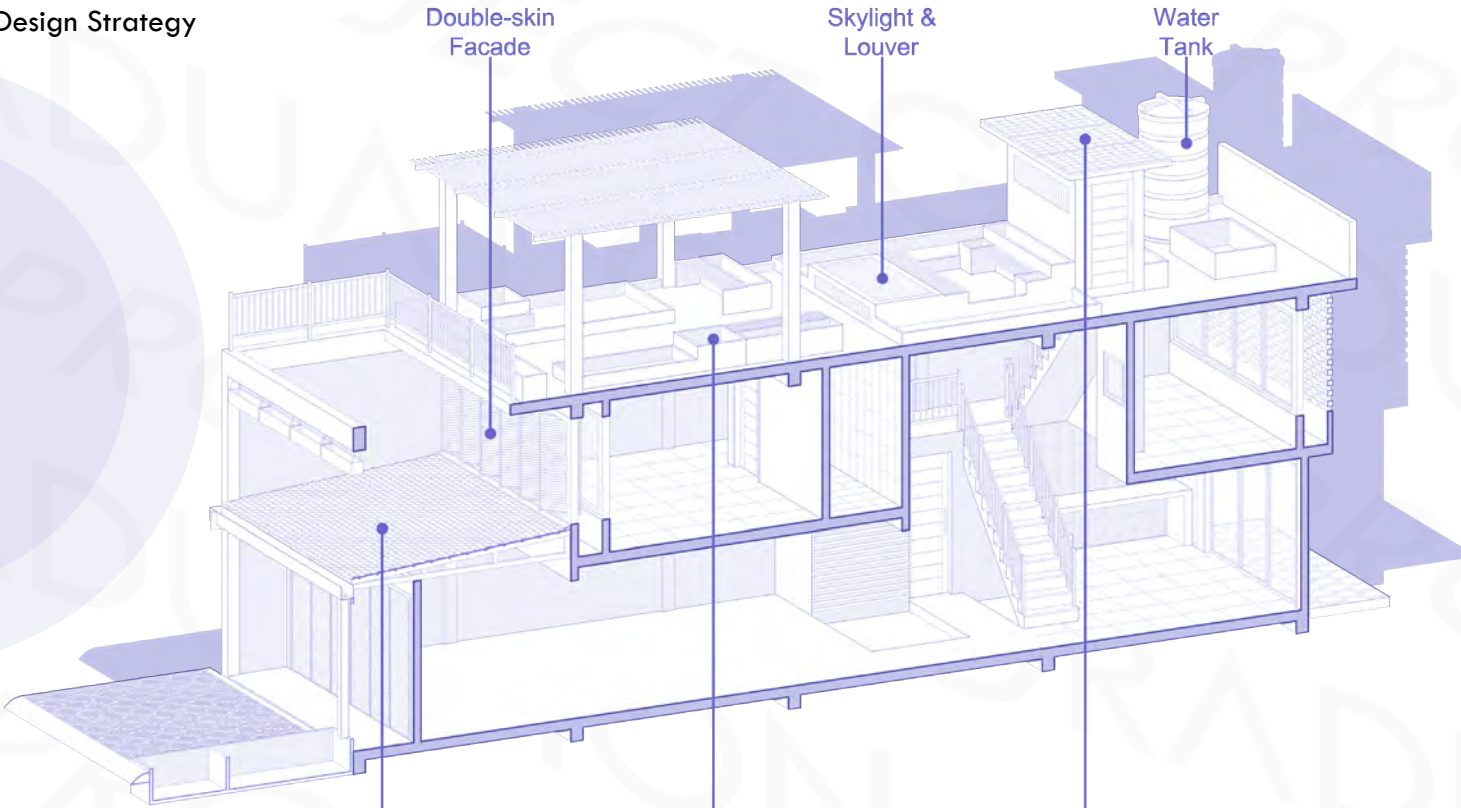
Determining Opening Specifications



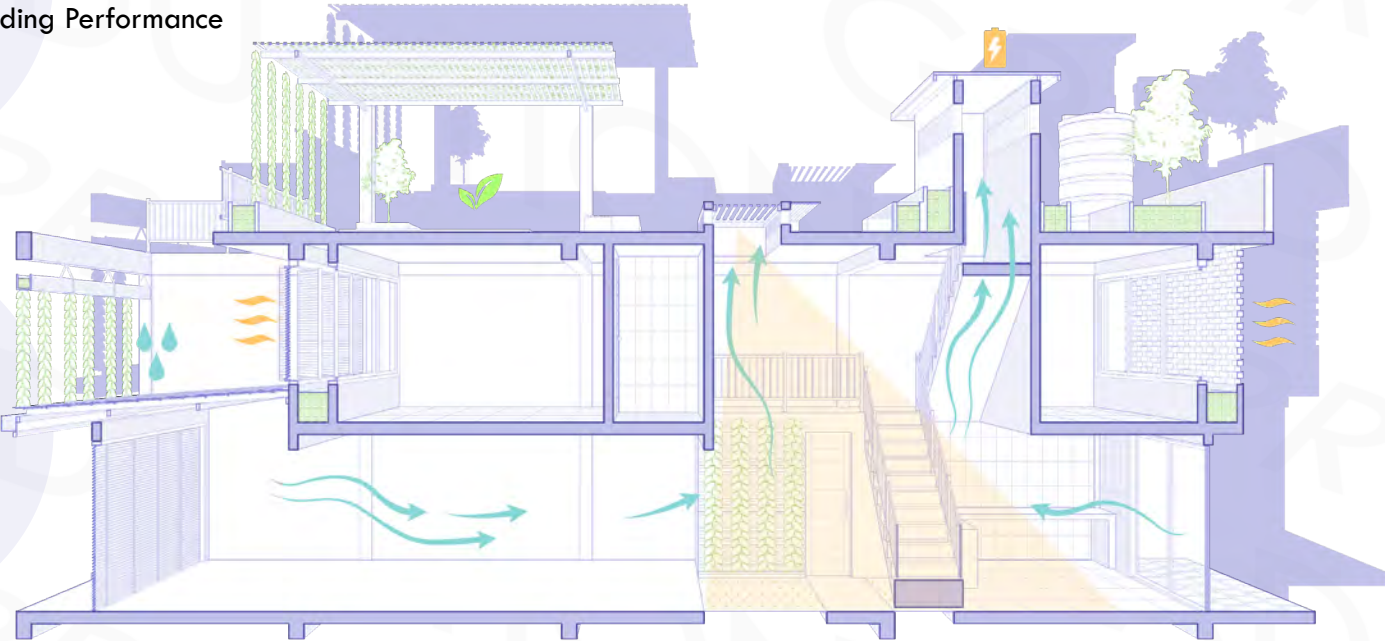
Utilizing Roof Space

The design transforms the existing building into a new form aligning with sustainable strategies. The considerations include the reduction of embodied carbon, betterment of building performance and living comfort, harvesting of natural resources, and compensation of built-up area.

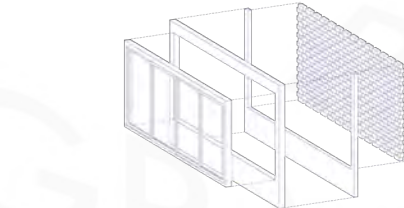
Design Strategy



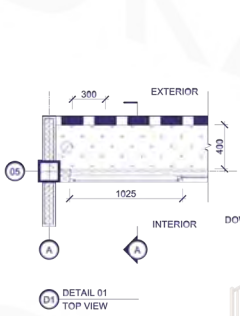
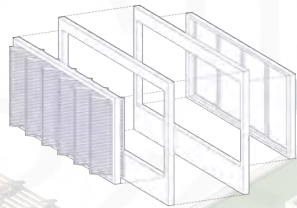
Building Performance



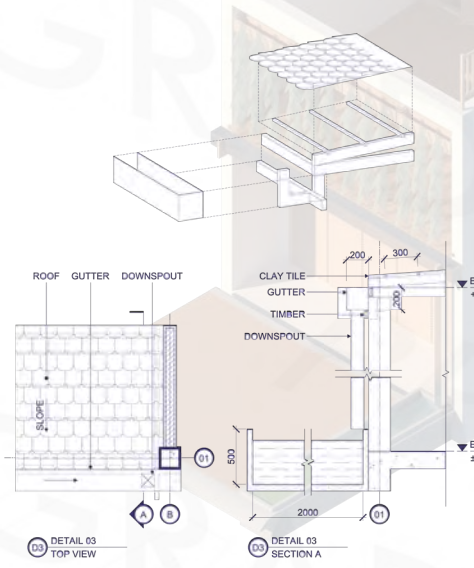
Rear Double-skin Facade



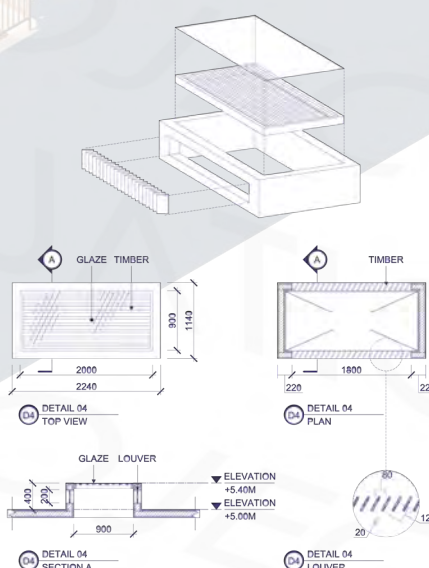
Front Double-skin Facade



Roof & Gutter



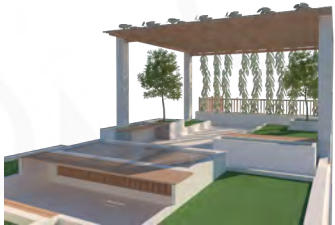
Skylight & Louver



Living & Working Space



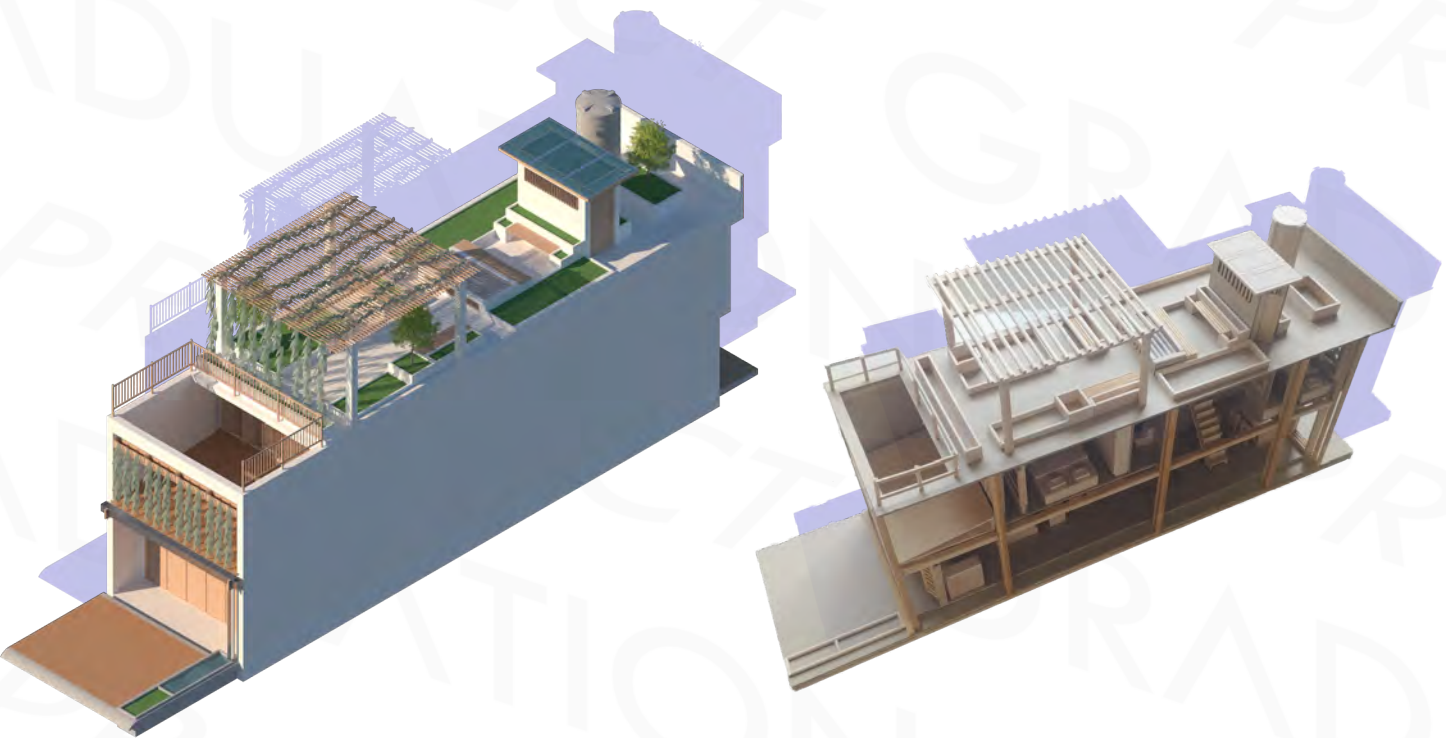
Bedroom

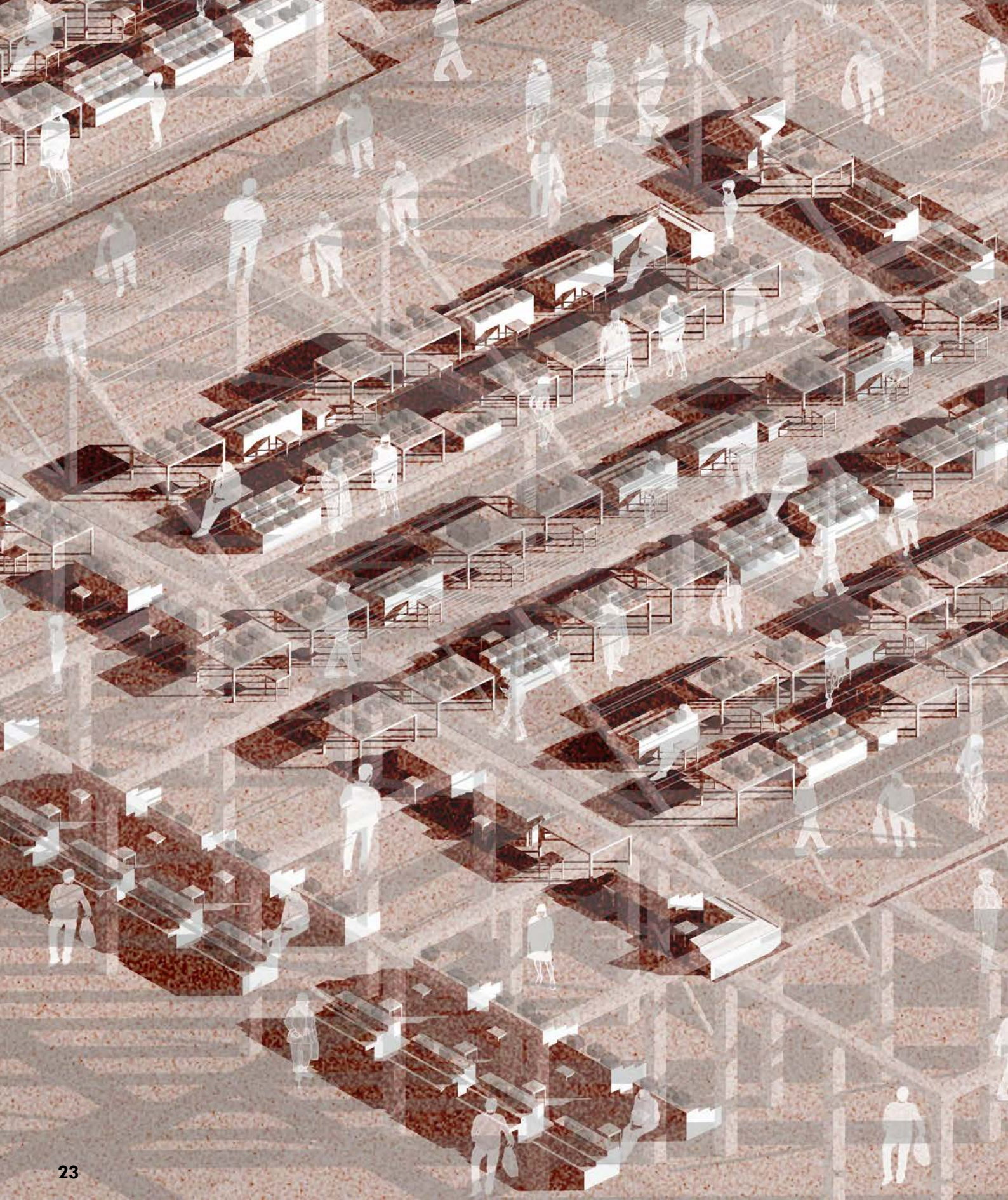


Roof Garden



The refurbishment design presents a new layout with the same functions as the existing building. The new ground floor arrangement is more open with better field of vision, circulation, and overall space functionality. The design of the mezzanine enhances aspects of comfort. The terrace roof is transformed into recreational area and greenery is integrated into every floor of the building. The renovation design expresses features of passive design, as well as the improvement of indoor environmental quality.





Beyond the Stalls

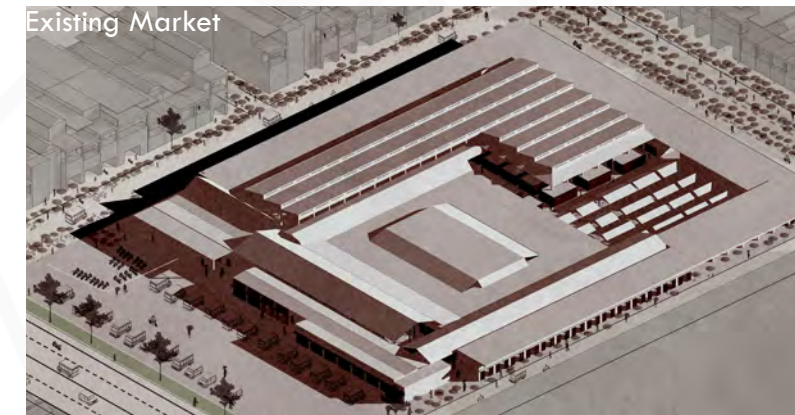
Reimagining the Spatial Use of Doeurm Kor Market

Lin Seka

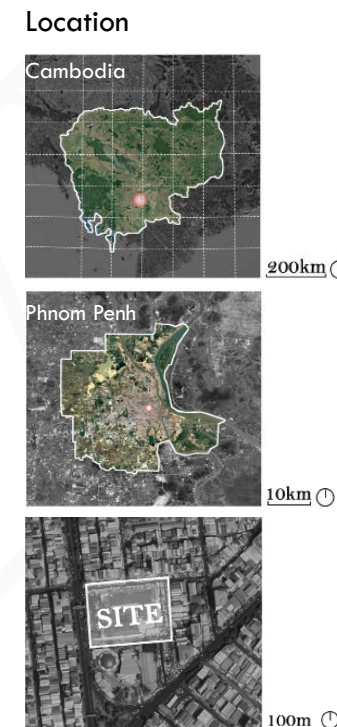
This study focuses on the redesign of the Doeurm Kor market in Phnom Penh using Jan Gehl's Cities for People theory, which emphasizes the concepts of protection, comfort, and delight.

Doeurm Kor market is a local wholesale market in Phnom Penh that serves as a central hub, connecting neighborhood vendors who supply products throughout the city. It operates 24 hours a day, seven days a week, selling a variety of goods ranging from fresh produce to accessories and dry products.

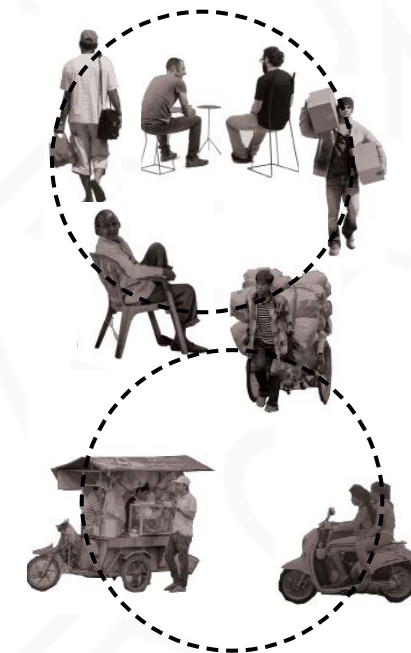
However, the market's current layout presents risks, including hygiene concerns from mixing wet and dry products, and traffic and weather dangers for roadside vendors. The proposed design aims to solve the problem by designing clearer circulation routes, rearranging zoning, and areas that promote walking, sitting, talking, and staying. Shading, open views, and social nodes such as food kiosks and play areas further enhance the user experience.



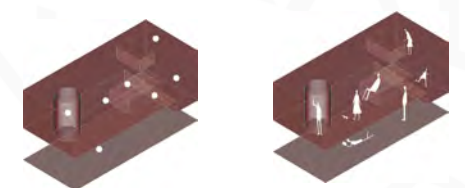
Existing Market



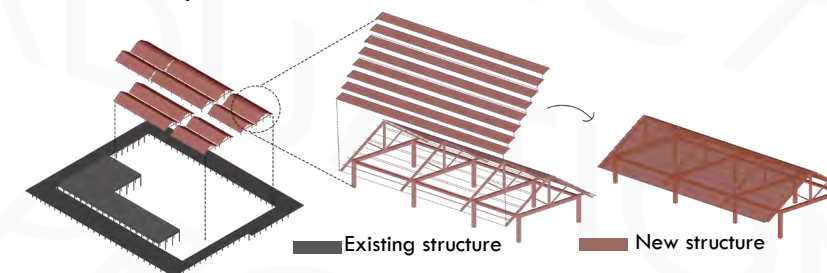
Activities

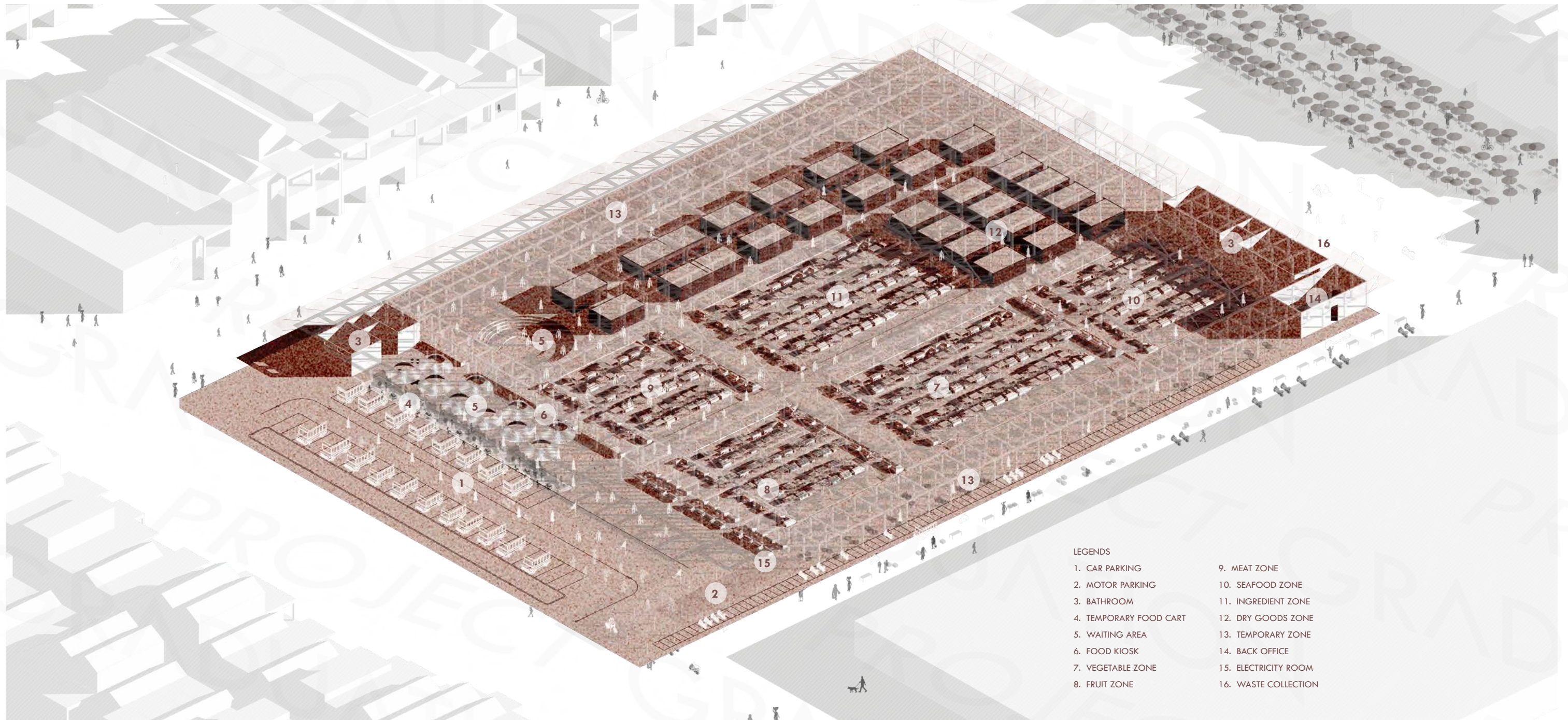


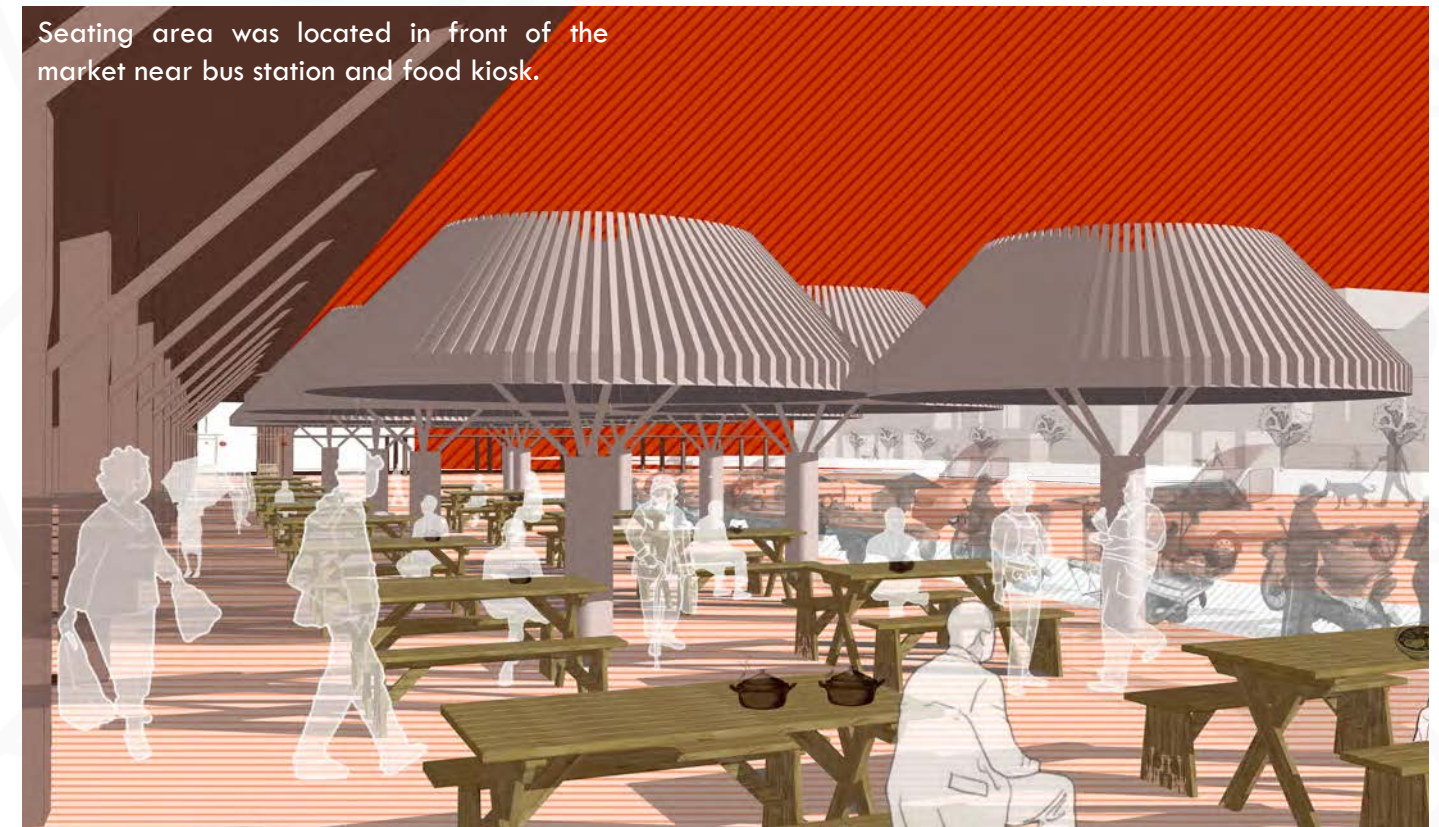
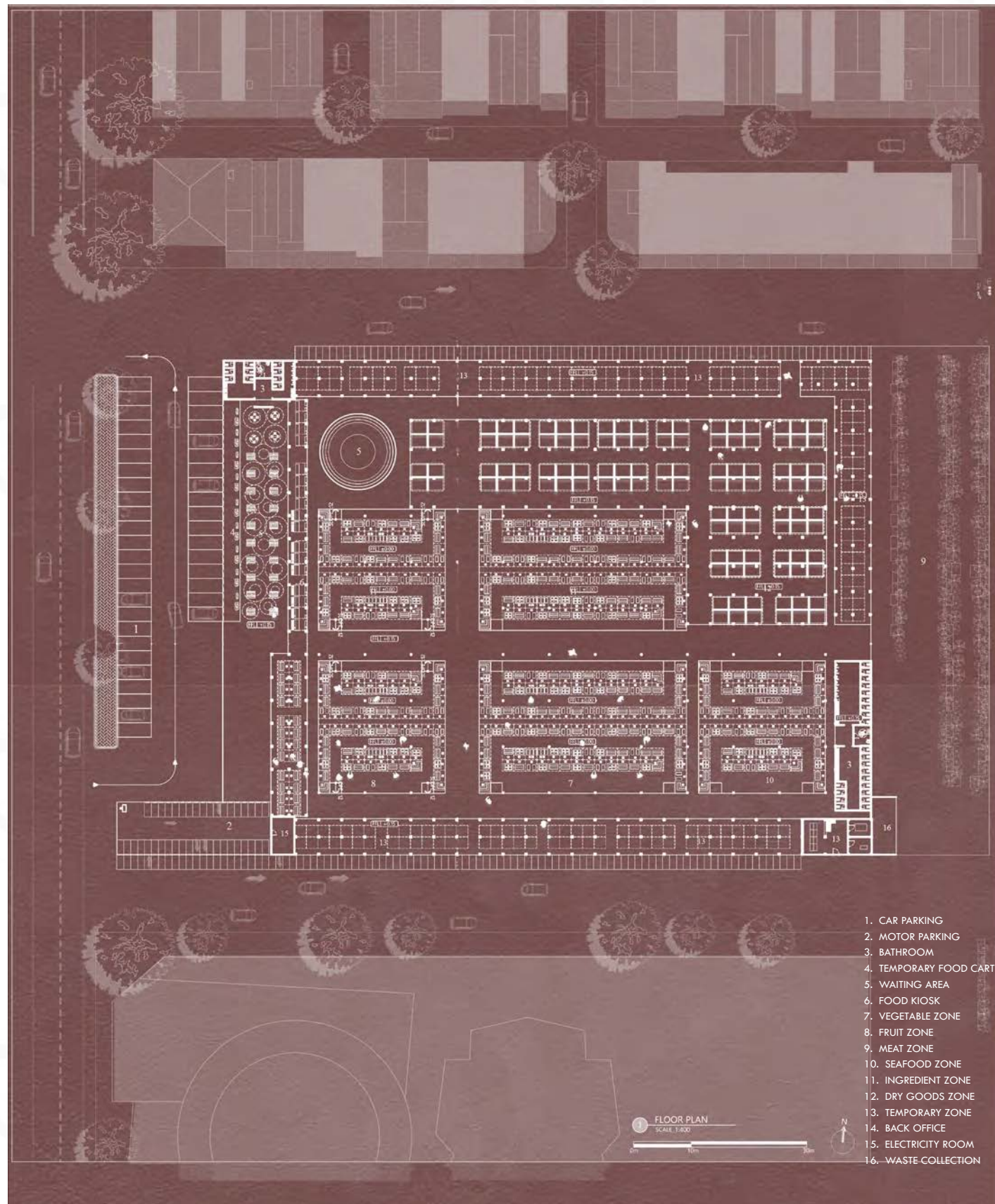
Concept Diagram



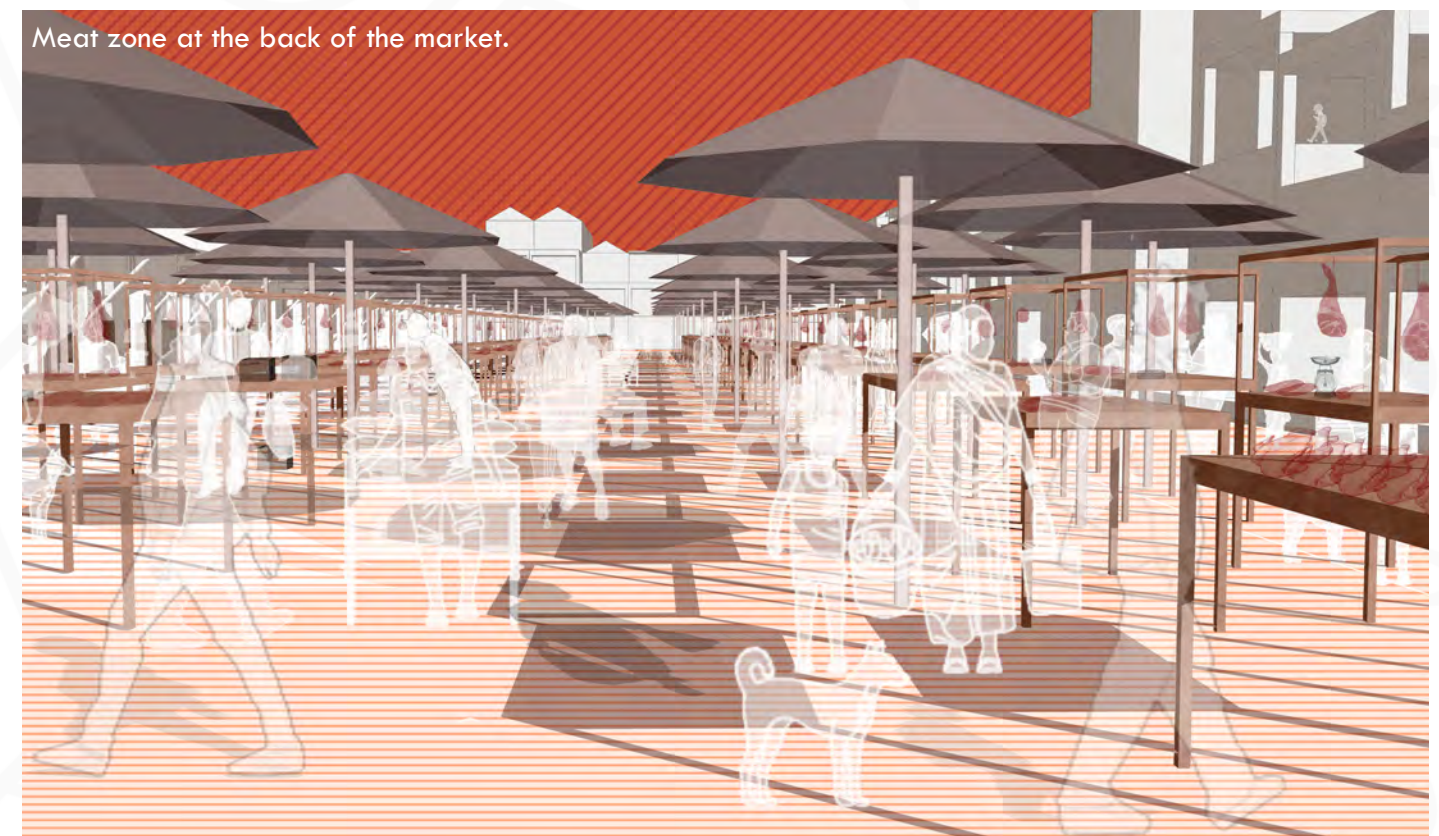
Form Development



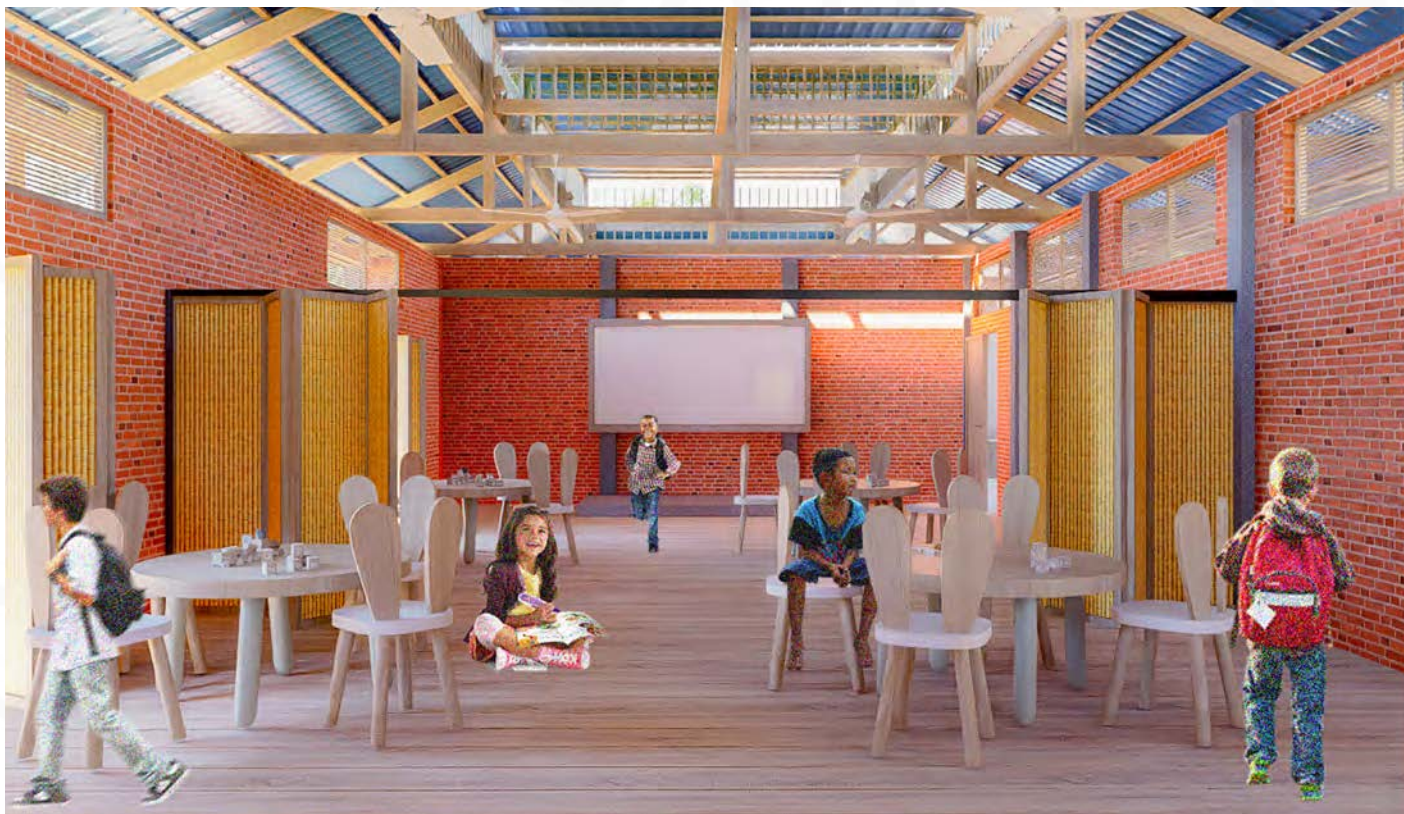




Seating area was located in front of the market near bus station and food kiosk.



Meat zone at the back of the market.

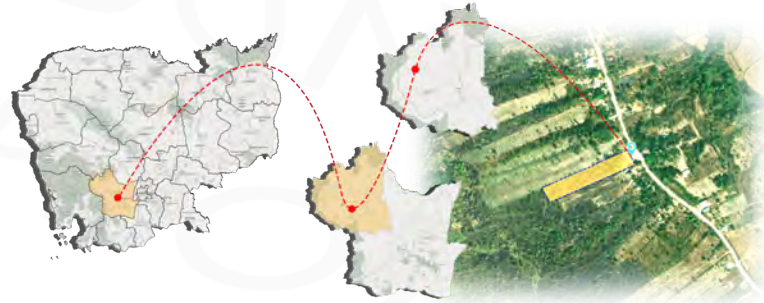


Rethinking Rural School Design for Self-Sufficiency

Phy Lymeng

Education in Cambodia has made notable progress in recent years, yet a strong gap remains between urban and rural learning environments. Many rural schools continue to operate with limited infrastructure, inadequate facilities, and learning spaces that do not support students' comfort or well-being. In many areas, schools also struggle with unstable electricity, and insufficient spaces for practical learning. These challenges directly affect students' concentration, attendance, and overall learning performance. Within this broader context, Oral Community School, located in rural Kampong Speu province, clearly reflects these issues. Established to serve children who previously had no access to formal education, the school operates with minimal resources and simple building structures. The conditions observed at Oral Community School highlight the urgent need for a more sustainable and resilient approach to rural educational infrastructure. This thesis responds to that need by proposing a school model that integrates self-sufficient systems and passive design strategies to enhance learning conditions while reducing dependence on external resources.

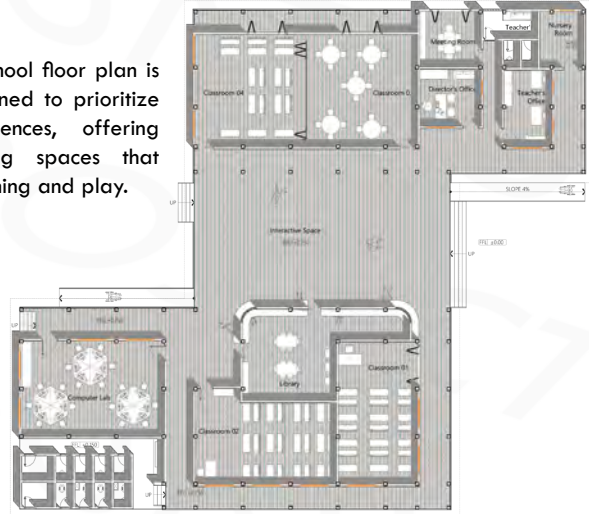
Location



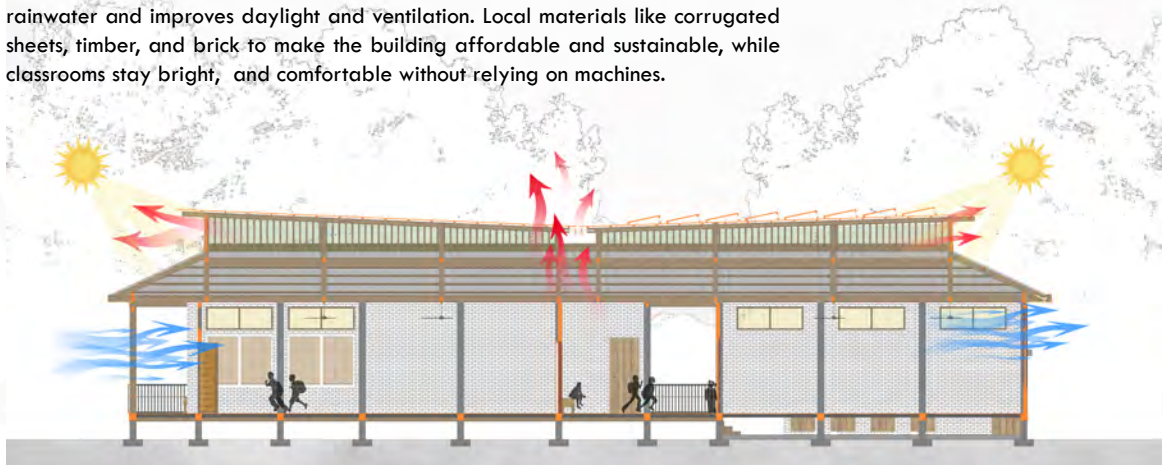
Phum Srae Ken, Trapeang Cho Commune, Oral District, Kampong Speu Province

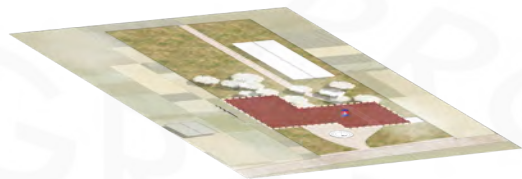
Floor Plan

This community school floor plan is thoughtfully designed to prioritize children's experiences, offering flexible, engaging spaces that support both learning and play.

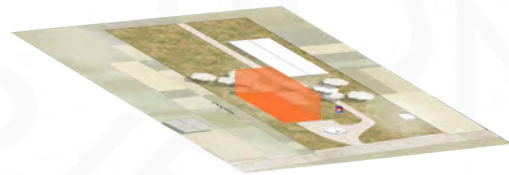


This section shows how the school uses passive design. The butterfly roof collects rainwater and improves daylight and ventilation. Local materials like corrugated sheets, timber, and brick to make the building affordable and sustainable, while classrooms stay bright, and comfortable without relying on machines.

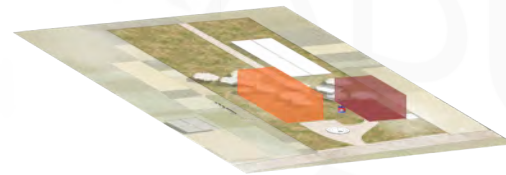




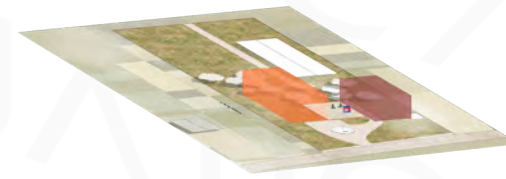
Setting the boundary for the school building within the existing masterplan, with consideration given to preserving the existing trees on site.



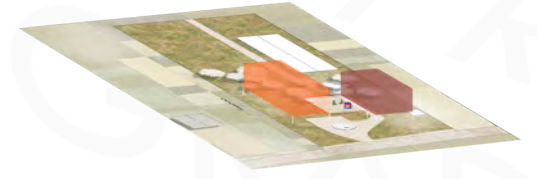
Starting with a single form.



Adding a new structure on the opposite side to provide additional functions.



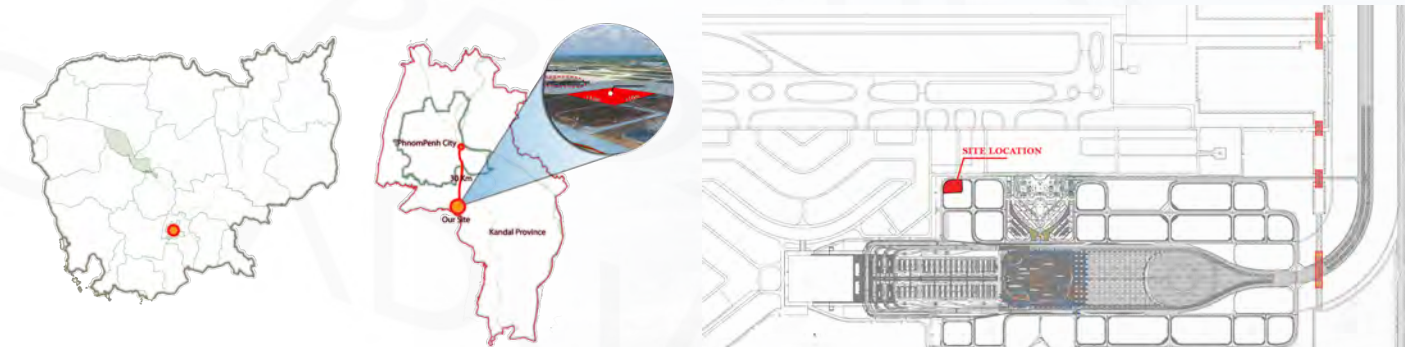
Creating an open space at the center of the site to support activities and community gatherings.



Raising up the entire structure above ground.



SITE LOCATION



The project is located at the center of the Techo international airport. It's located right next to the main terminal building around 400-500m, the VVIP boarding building and the main parking area.

MASTER PLAN

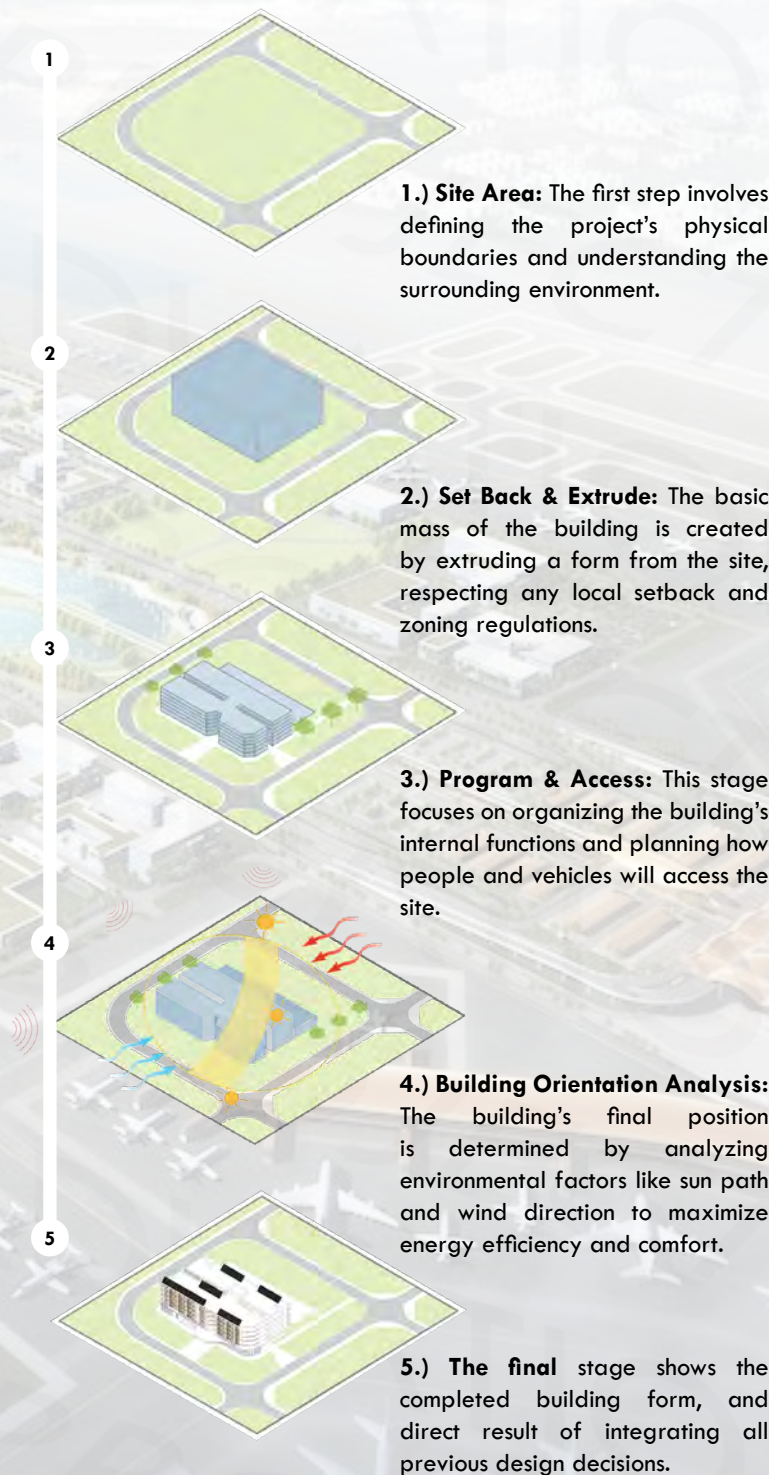


Merging Heritage: Rethinking Airport Hotels for Development Economies in Cambodia

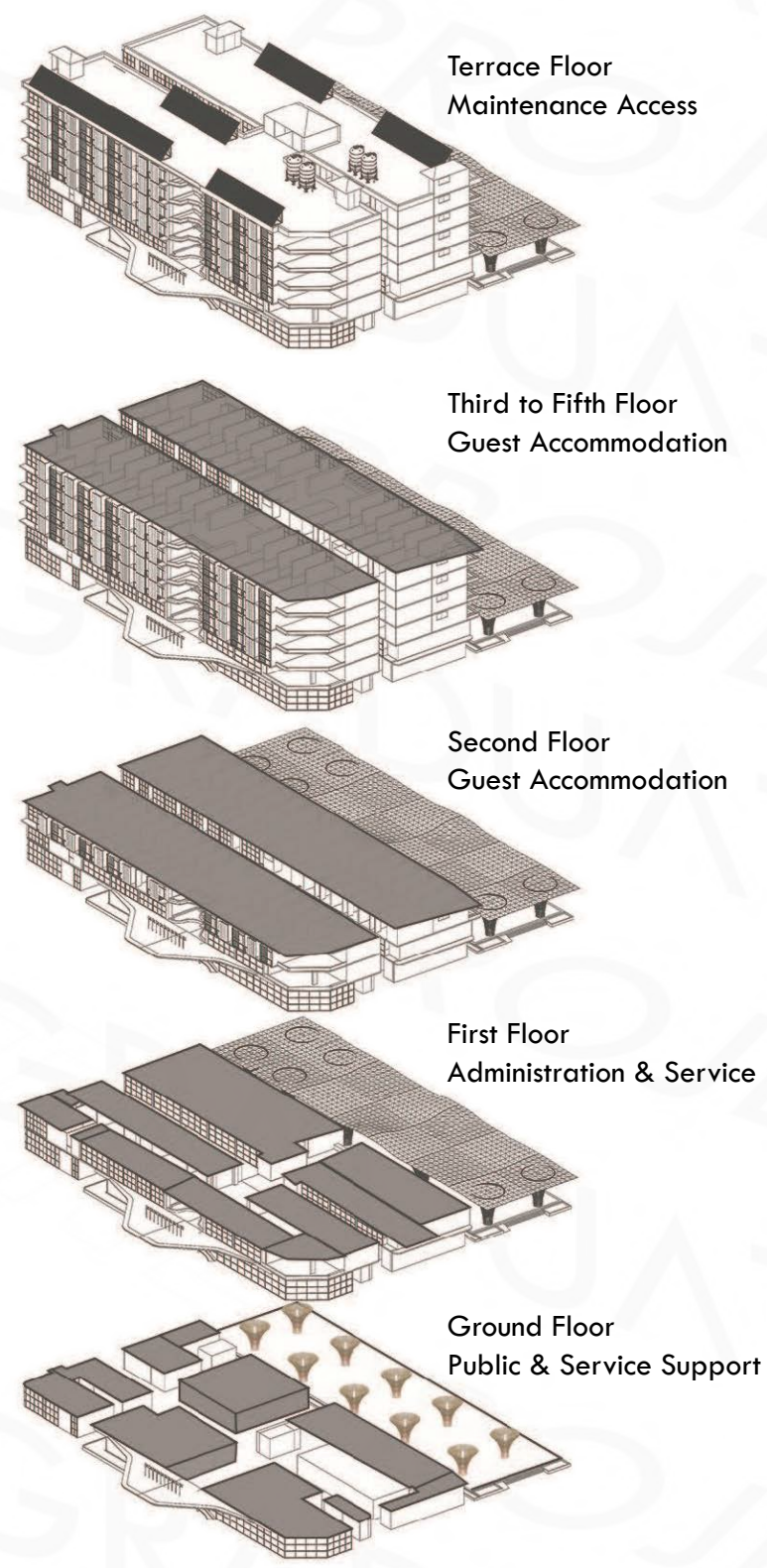
Te Senghok

The project “Merging heritage: rethinking airport hotel for development economies in Cambodia” by integrating Khmer cultural identity with modern sustainable design. Through the integration of adaptive reuse principles, ergonomic planning, placemaking, and bioclimatic strategies, the design demonstrates how cultural heritage can coexist with functionality and environmental performance. The proposed 4-star airport hotel at Techo International Airport (TIA) aims to serve not only as a place of rest for travelers but also as a cultural gateway that reflects Cambodia’s hospitality, identity, and forward-looking vision. It contributes to economic growth, enhances tourism potential, and promotes sustainable design practices within the built environment.

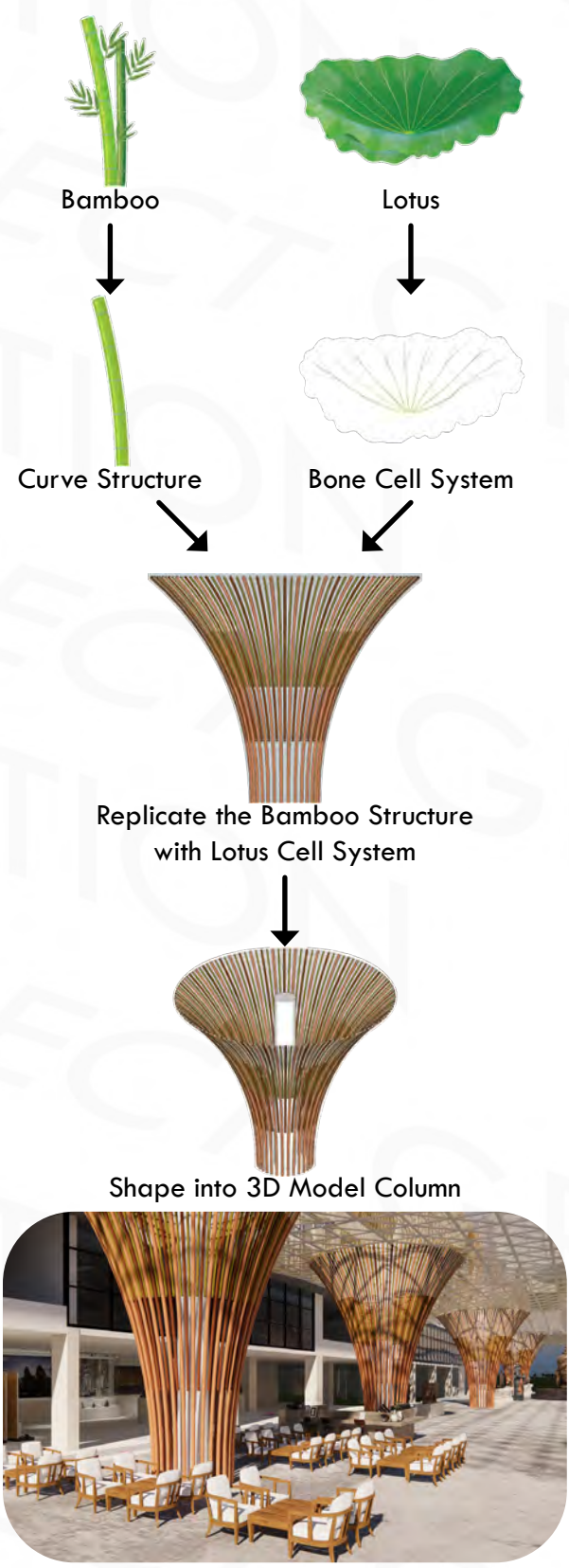
FORM DEVELOPMENT



ILLUSTRATION



PAVILION FORM DEVELOPMENT



VISUALIZATION



The design concept is a fusion of modern sustainable architecture and Khmer culture. By incorporating Khmer ornamentation and cultural materials, the hotel will serve as a cultural ambassador for Cambodia, offering visitors a unique and lasting experience beyond standard accommodation.

Lobby & Waiting Area



An immersive space that blends modern design with traditional Khmer artistry. The design showcases detailed murals, statues, and rich textures, all inspired by classic Khmer motifs, creating a serene of culturally rich welcome for guests.

Khmer Galleries



This space is design for guest to engage with Cambodian heritage and invite guest to explore and appreciate a collection of sculptures and legacy of Khmer people.



Standard Double Room



Standard Twin Room



Cafeteria



Galleries

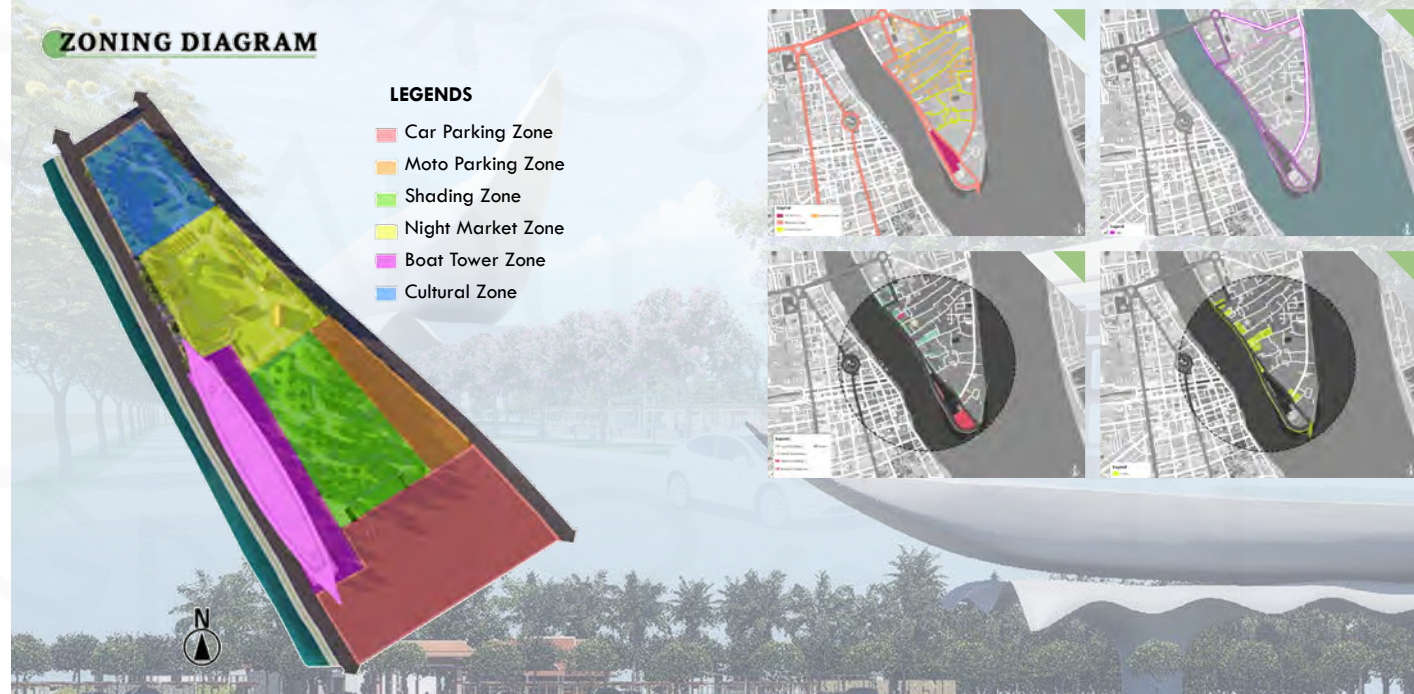
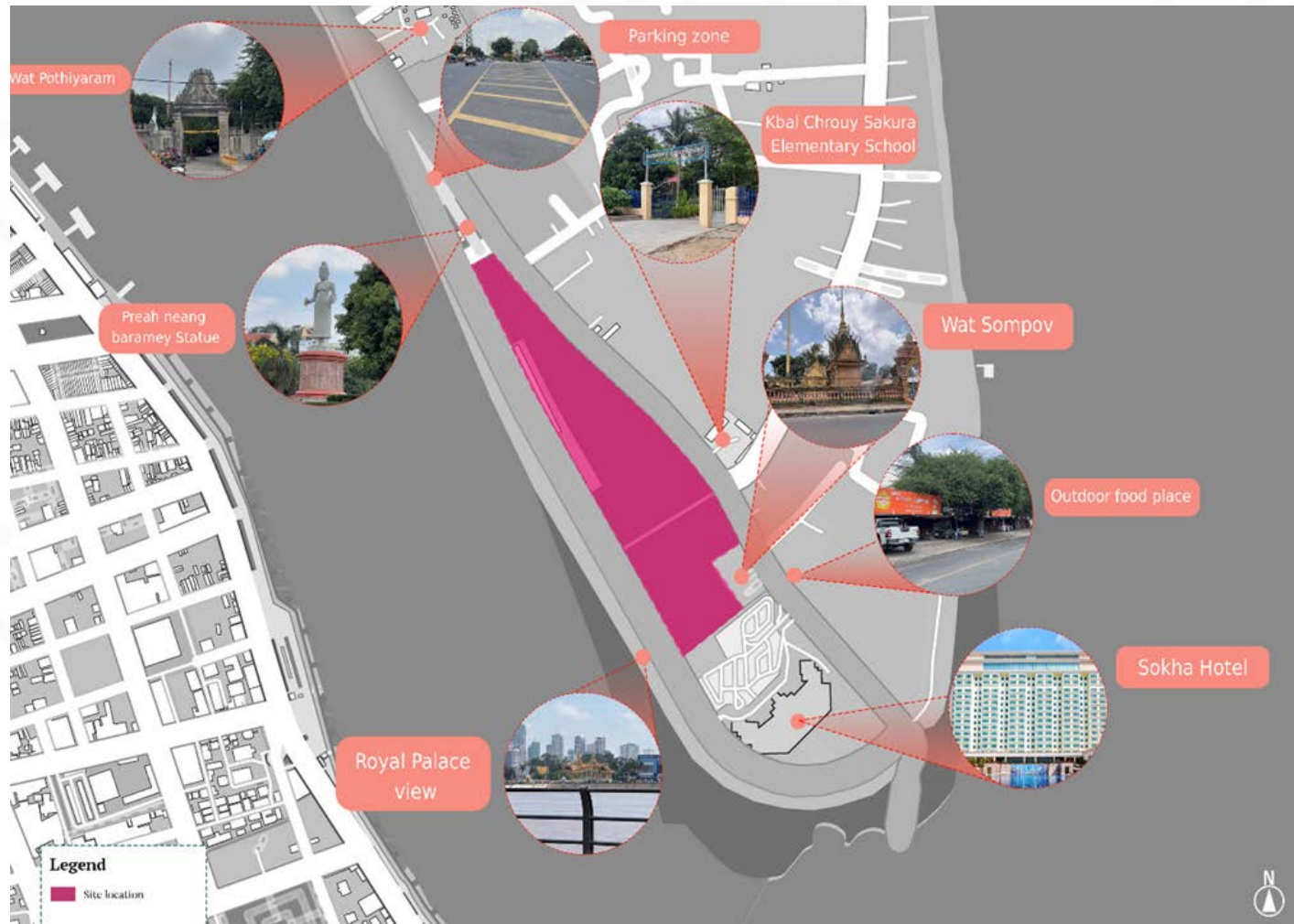


Kiosk



Swimming Pool





The Case Study of Riverside Land Integration Public Park Proposal at Chroy Changvar

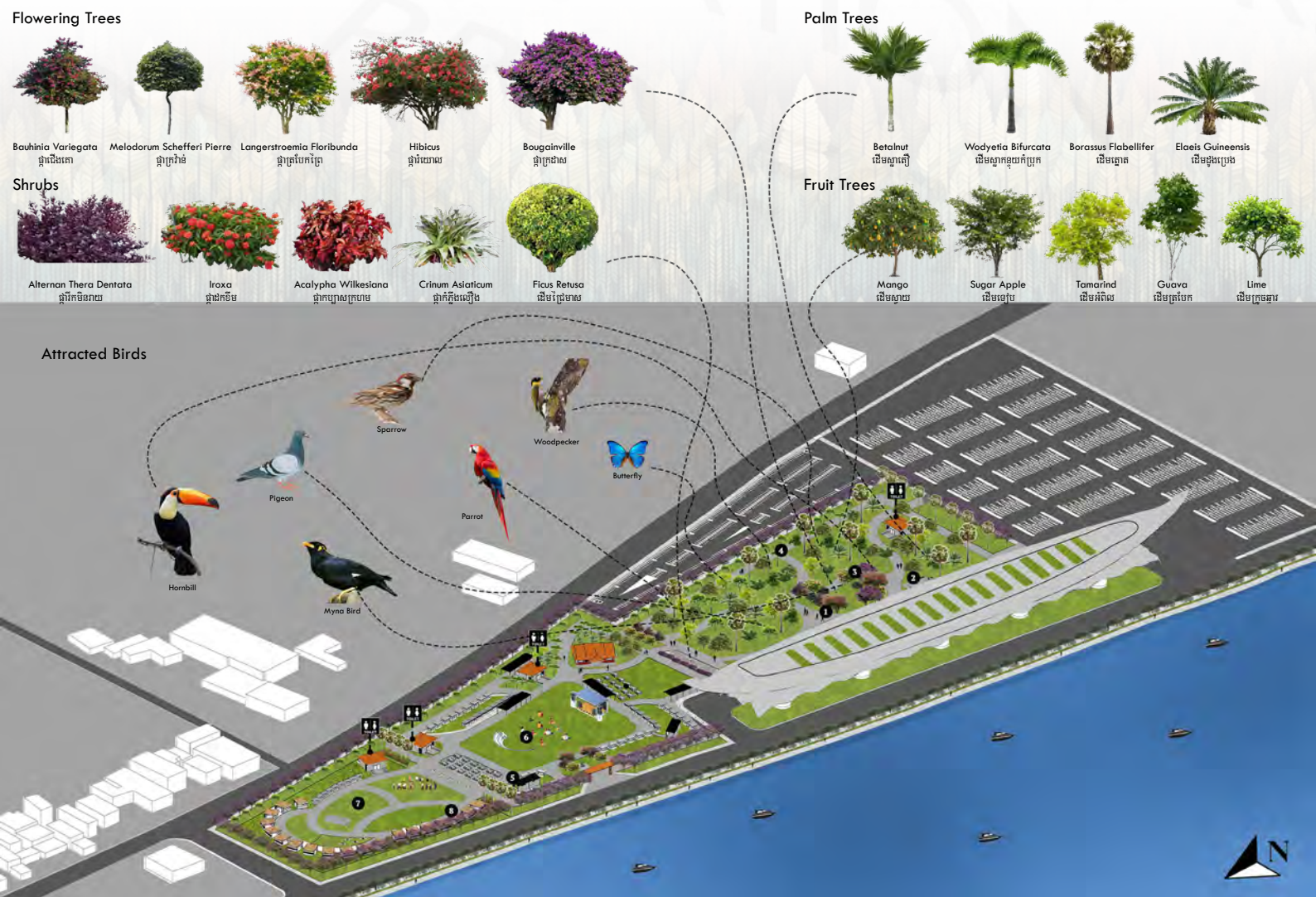
Iary Yuttpul



The project transforms approximately 57,000 sqm of vacant land along the Chroy Changvar riverside into a multifunctional public park that reconnects people to the Tonle Sap River and contributes to the revitalization of Phnom Penh's riverfront. The site has been fenced and inactive for over a decade, creating a visual and social gap in the city's landscape. To address this, the design introduces a riverside walk, cultural zone, and night market that promote community activity, cultural exchange, and environmental sustainability. Guide by Kevin Lynch and Jane Jacob's urban design theories, the project emphasizes accessibility, safety, and mixed use, creating a vibrant, inclusive, and culturally expressive public space that support local economic activity.



Perspective of Masterplan



Each zone is scaled to match its function and user needs, open areas for events, compact zones for vendors, and shaded pockets for rest. The layout balances activity and quiet, ensuring no space feels overcrowded or underused.

The park is designed to serve a wide range of age groups. Each zone offers tailored experiences: shaded seating and gentle paths for older visitors, creative and active spaces for youth, and flexible areas for socializing, dining, and cultural engagement across generations.

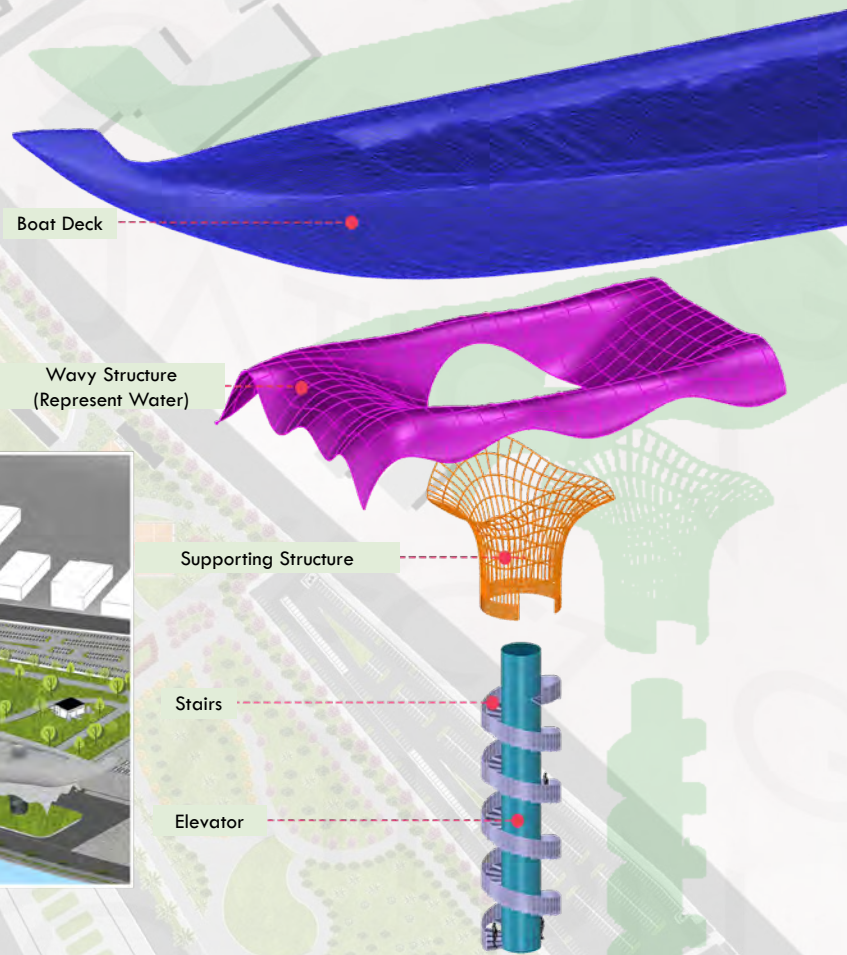


The park also has concept inspiration from the site, as it is surrounded by a river, such as in the greenery zone is a concept shape representing the land of Chroy Changvar at the center and where Tonle Sap and Mekong River meet and the zone has a central land where people can walk on the path but still meet up at the center same way as both rivers meet.

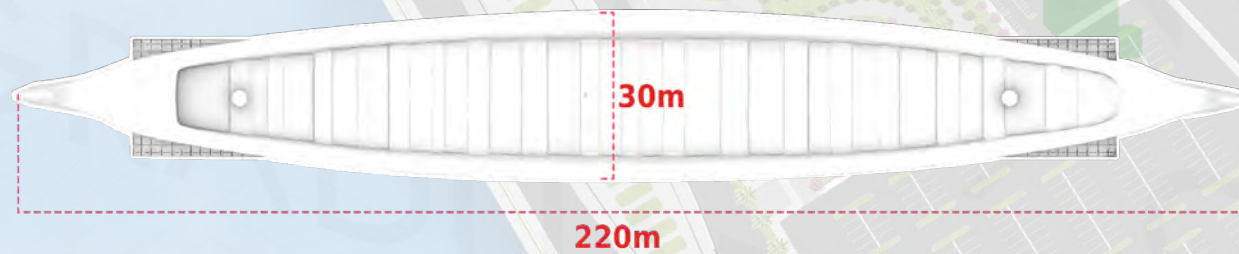
LANDMARK

The pavilion connects visually to the Royal Palace and takes inspiration from the Bon Om Touk Water Festival, with a boat-like form symbolizing both celebration and Angkor's naval heritage. Elevated on wave-shaped base, it stands as a landmark visible from the palace, glowing at night like a boat floating in the air. Beyond its symbolic role, it may serve as a skybar with a \$5 entry fee, offering drinks and panoramic festival views.

EXPLODE VIEW



TOP VIEW

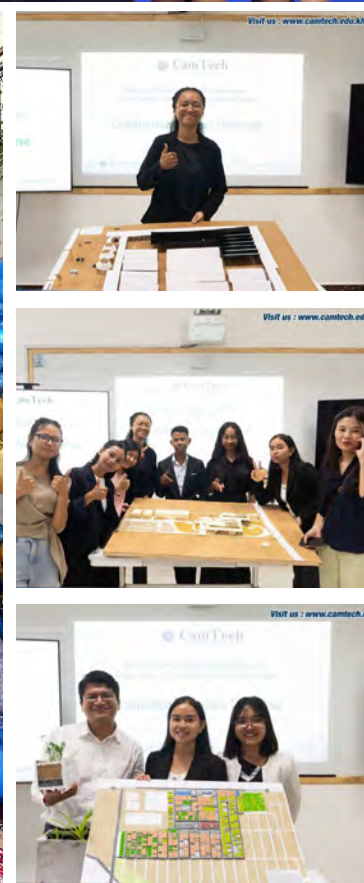


FRONT VIEW



Visitors begin at the shaded entrance, journey through the cultural zone, and return via the night market, creating a walkable loop that blends comfort, activity, and cultural engagement.





DEPARTMENT OF ARCHITECTURE AND INTERIOR DESIGN GRADUATION PROJECT 2025

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